



Type SHGV SF6 Gas Insulation Vacuum OLTC Installation and Operation Instruction

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Shanghai Huaming Power Equipment Co., Ltd.

Safety Notice

1. Please read this instruction thoroughly before using the equipment.
2. The maintenance of SHGV OLTC must be carried out by specially trained technicians.
3. With products' upgrade and improvement, Huaming reserves the right to change technical specifications, product manuals and technical data provided.
4. For products whose technical data are beyond this instruction's specification, and for applications in special occasions, you may contact Huaming to customize special designs. This instruction includes all information required for the operation and installation of SHGV OLTC.
5. Do not assemble or disassemble tap-changers without Huaming's authorization.
6. Do not let tap-changers run overload for a long time.
7. The OLTC electrical part can be operated by a professional electrical engineer.
8. Do not lift the insert and check the tap-changer on wet and snowy days.
9. Before the transformer being put into use, make sure the tap-changer is securely grounded to earth.
10. Before operation, make sure the tap-changer and the motor drive unit are at the same working position.
11. Do not screw the bolts on the tap-changer head flange.

	Notice	This technical material and other applicable documents must be kept at hand for future use.
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Safety Information and Regulations

Please pay extra attention to such words as Attention, Notice, Warning! and so on in this instruction, so as to ensure the safety of on-site people and the tap-changer during installation, operation and adjustment.

Please obey this instruction's premises and requirements and read the notices and those on the package or the product to avoid any possible danger. This regulation and information apply to all stages within life expectancy from the product delivery, installation, and operation to disassemble and scrap.

Huaming strictly adheres to IEC 60214-1 & GB10230.1 standards as well as the company's own quality regulation system to ensure a steady high-quality craft. So in the premise of correct use and obedience of all relevant technical material, the use of this product is safe to human beings and the environment.



Attention (Notice)



Danger of tipping



Electric Shock


Warning
(Danger)


Fire Hazard

Warning!

If the sealing gas inside the transformer leaks, you must stay away from the transformer room!



- If there is gas left, people might suffocate due to lack of oxygen.
- Keep the transformer room ventilated.
- SF6 is heavy gas, so there is some remaining in hollow places.
- Before entering the transformer room, make sure the oxygen concentration is higher than 18%.

Warning!

Do not breathe in SF6 when there is a transformer malfunction.



- SF6 may become poisonous in case of a transformer malfunction.
- If breathing it in, workers may get poisoned.
- It must be collected in a sealing container.

Warning!

Bolts that keep the tap-changer gas-tight must not be unscrewed.



- Otherwise, reduced pressure may lead to malfunctions, and the leaking gas may pollute the environment!

CONTENTS

Safety Notice	1
Safety Information and Regulations.....	1
1. General.....	5
1.1 Functions.....	5
1.2 Technical specifications.....	5
1.3 Product structure.....	6
2. Packaging, Transport and Storage.....	7
2.1 Parts delivered.....	7
2.2 Packaging and acceptance.....	7
2.2.1 Transport.....	8
2.2.2 Receipt.....	8
2.2.3 Unpacking and checking.....	9
2.3 Storage.....	10
3. OLTC Installation in the transformer.....	11
3.1 Flange on the transformer house used to fasten the tap-changer.....	11
3.2 OLTC installation in the bell-type transformer.....	12
3.2.1 Tap-changer head flange disassembly and the switch insert lifting.....	13
3.2.2 Lifting out installation flange	15
3.2.3 Assembly of the tap selector and diverter switch gas compartment.....	16
3.2.4 Install OLTC into bell type transformer tank.....	20
3.2.5 Install bell type transformer tank cover.....	20
3.2.6 Install diverter switch insert.....	21
3.3 OLTC installation in the standard-type transformer	23
3.3.1 Disassemble OLTC head installation flange and lift up diverter switch insert ☐ Refer to 3.2.1.....	23
3.3.2 Lift out installation flange ☐ Refer to 3.2.2	23
3.3.3 Assemble tap selector and diverter switch gas compartment ☐ Refer to 3.2.3	23
3.4 Assemble installation flange on the transformer cover.....	23
3.4.1 Raise the SHGV by lifting device to the mounting surface of the bell flange supporting flange for connection.....	23
3.4.2 Installation methods of diverter switch insert and head cover are same as that of the bell type.....	23
3.5 Connection of tap winding leads and OLTC connection terminals.....	23
3.5.1 Fixation of tap winding leads and tap changer connection terminals..	24
3.5.2 Tap selector terminal leads shall not cause forced distortion or damage to tap selector.....	25
4. Transformer ratio test and DC resistance measuring.....	25
4.1 Transformer ratio test (mount temporarily head flange & diverter switch insert for bell type installation)	25
4.2 Measuring DC resistance on transformer.....	26
5. Drying process.....	27
5.1 Vacuum drying	28
5.1.1 Vacuum drying in autoclave	28
5.1.2 Vacuum drying in transformer tank	28
5.1.3 Treatment on OLTC after drying process	28
6. Installing protective device and drive mechanism accessories	29
6.1 Installation of rapid pressure rise sensor	29
6.2 Installation of motor drive unit	29
6.3 Installation of bevel gearbox	29
6.4 Installation of horizontal and vertical shafts.....	31
6.5 Installation of protective cover.....	33
6.5.1 Installation of horizontal protective cover.....	33
6.5.2 Installation of vertical protective cover (Figure 26).....	34
6.6 Adjustment position of OLTC drive system.....	35

7. SF6 gas filling	36
8. OLTC type test and preparation before test.....	37
8.1 Preparation before test.....	37
8.1.1 OLTC pressure inspection	37
8.1.2 Grounding connections	37
8.2 Test in transformer factory.....	38
8.2.1 Operation test.....	38
8.2.2 Electric test of transformer	38
9. Inspection before operation.....	39
9.1 Visual inspection	39
9.2 Inspection of manual operation.....	39
9.3 Inspection of electric operation.....	39
9.4 Transformer ratio measurement test	39
10. OLTC operation monitoring and maintenance.....	40
10.1 Operation monitoring	40
10.2 The maintenance of OLTC.....	40
10.3 Routing inspection	40
10.4 Periodic inspection	40
Appendix.....	41

1. General

1.1 Functions

In response to the increasing number of SF6 gas transformers put into operation in China, Huaming designed the SHGV gas insulated vacuum OLTC based on its original vacuum type.

SHGV gas-insulated vacuum OLTC (hereinafter referred to as OLTC) uses vacuum interrupters instead of arcing contacts, which has prominent advantages:

1. The current breaking is in the vacuum interrupters and the arc created is extinguished in it, causing no influence on the gas properties of SF6.
2. The task of carrying current for a long time is undertaken by special mechanical main contacts, while the vacuum interrupters only take the current instantly through, so the overload capacity of OLTC has been greatly improved.
3. All the contacts use rolling contact.

1.2 Technical specifications

NO.	Items		Parameter		
1	Max. rated through current (A)		400		
2	Max. rated step voltage (V)		4000		
3	Rated step capacity		1200		
4	Rated frequency (Hz)		50 or 60		
5	Connection		3-phase Y-connection for neutral point/ Single-phase for any selectable winding connection		
6	Short-circuit current test (kA)	Thermal (3s)	6		
7		Dynamic (peak)	15		
8	Insulation to ground	The highest voltage for equipment (kV)	40.5	72.5	126
9		Rated separate source AC withstand voltage(kV/50Hz,1min)	95	140	230
		Rated lightning impulse withstand voltage (kV,1.2/50μs)	225	350	550
10	Max. operating positions		27		
11	Mechanical life		Not less than 1,500,000 operations		
12	Electrical life		Not less than 300,000 operations		
13	Gas compartment of diverter switch	Service pressure	0.025 MPa		
		Leakage test	No leakage under 0.08 MPa for 24 hours		
		Under pressure protection	0.015 MPa (Density sensor alarm)		
		Over pressure protection	0.2 MPa (Trip of emergency pressure relay)		
14	Motor drive unit		SHM-D		

1.3 Product Structure

SHGV is the combined type and is composed of diverter switch and tap selector. (See Figure1)

SHGV is installed on the top cover of transformer by its head flange, and connected to the motor drive unit through the upper gear box on the cover of OLTC, bevel gear (auxiliary devices) and driving shaft (horizontal & vertical), realizing the switching operation of OLTC by motor or remote control.

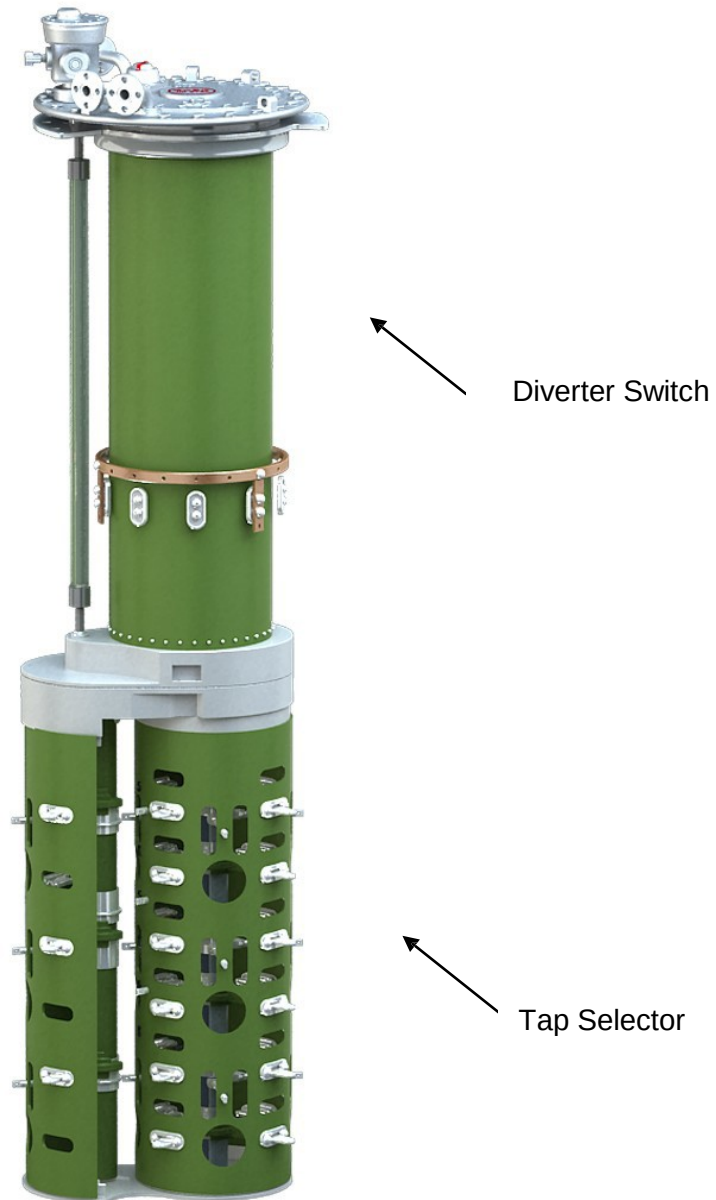


Figure 1 SHGV OLTC

2. Packaging, Transport and Storage

2.1 Parts delivered

The whole OLTC package delivered is composed of following parts:

- diverter switch
- tap selector
- motor drive unit, controller and connecting cable
- transmission shaft, coupling part, bevel gear box and other transmission accessories
- protective relay and its accessories
- special tools and accessories to users for tap-changer assembly
- special components required by users

For actual parts delivered, please refer to the packing list shipped with the product.

2.2 Packaging and acceptance

Warning!	
Danger of death and severe injury!	
	Danger due to tipping or falling load. ➤ Only trained and appointed people may select the sling gear and secure the load. ➤ Do not walk under the hanging load. ➤ Use means of transport and hoists with a carrying capacity of > 1000 kg.

Warning!	
Damage to property!	
	Risk of damage to the on-load tap-changer resulting from falling and tipping over! ➤ Only trained and appointed people may select the sling gear and secure the load. ➤ Do not walk under the hanging load. ➤ Use means of transport and hoists with a carrying capacity of > 1000 kg.

2.2.1 Transport

1. Before sent out from the factory, the diverter switch should be stored in a separate iron can which is filled with nitrogen. During the transportation, do not stack anything on the iron can.
2. The tap selector and accessories are packed in a stable wooden crate, suitable for various transportation means. They are allowed to be stacked properly, but the weight on the crate top must not exceed 500kg/m².
3. The packages should be placed properly based on the position of center gravity. When hoisting the packages, please follow up the pictograms and do the operation based on standards.
4. Shipping pictograms (See Figure2)



Figure 2 Shipping Pictograms

5. OLTC moving parts and contacts have been lubricated. Note that grease is not removed and care must be taken to avoid dust and other particulate matter.

2.2.2 Receipt

Users should pay attention to this before accepting the product: every delivered shipment must be checked for the following items by the recipient before acceptance (acknowledgment of receipt):

- Completeness based on the delivery slip
- External damage of any type

If external transport damage is detected on receipt of the shipment (if possible, immediately take a picture of the package and the damaged product), or if the product does not meet with the packing list, do not accept the package. Instead, talk to the transportation company right away. If no results are reached, please email Huaming After-Sales Department.

The above procedure also applies to packed product's corrosion caused by moisture (rain, snow and dew).

The parts must be stored in a dry place before installation.

- OLTC must be sealed in the package. Only open before installation.

	Notice	➤ The transportation and lifting of the package must be done by special technician. ➤ Use means of transport and hoists with a carrying capacity of > 1000 kg.
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Warning!	
	Danger of death and severe injury!
	Danger due to tipping or falling load. ➤ Only trained and appointed people may select the sling gear and secure the load. ➤ Do not walk under the hanging load. ➤ Use means of transport and hoists with a carrying capacity of > 1000 kg.

The packing box should be firmly fixed and be handled with care during shipping and lifting to avoid the possibility of damages due to vibration, crashes, dumping, bumps and sloshing.

If the box crashes or gets sever impact, damages are possible. Conduct a comprehensive inspection. If necessary, the inspection or maintenance must be done by professional technicians from the manufacturer.

2.2.3 Unpacking and checking

Warning!	
	Damage to property!
	Risk of damage to the on-load tap-changer resulting from falling and tipping over! ➤ If the sealing package is damaged, do not assemble or put the packed gas compartment/switch insert into operation. ➤ Dry the gas compartment/switch insert according to the operation instruction or send it back to Huaming for drying.

Unpacking and transportation damage check:

- Send the package to a safety place
- Check the product and accessories based on the packing list

	Notice	When unpacking, avoid damaging the original package.
		If the product and accessories are not in line with the packing list, immediately contact the manufacturer.

2.3 Storage

When the packed product is stored for more than one year, it should be carefully examined before the installation. Special parts with sealed packaging can be stored outdoors if the following conditions are met.

Following requirements must be met when choosing and putting up a storage place:

- During operation, OLTC's operating temperature must be above -25°C and not exceed 105°C.
- The ambient temperature must be above -25°C and not exceed 40 °C.
- The storage place must be free from flammable, explosive and corrosive gas.
- The product must be protected from dangers like moisture, dust, mice, ants, etc.
- Inspect the product regularly in case of any incident.

If the product is stored for a long time, the desiccant must be replaced regularly and a sealed package should be applied.

3. OLTC installation in the transformer

3.1 Flange on the transformer house used to fasten the tap-changer

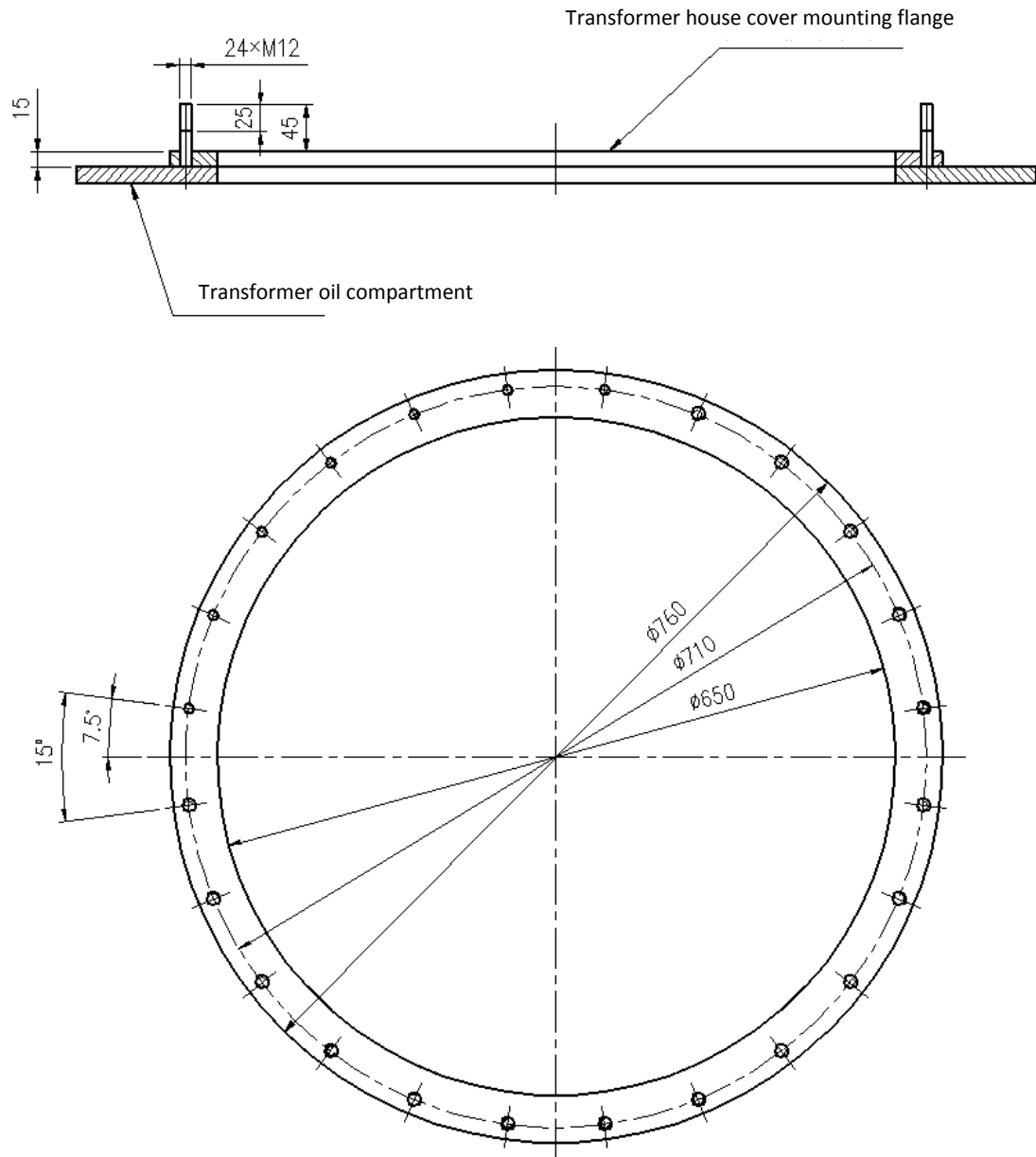


Figure 3 Flange used to fix OLTC

Mounting flange is needed to fix OLTC head to the transformer cover. The flange structure should be designed according to the shape of OLTC head's seal surface (See Figure3). The double-headed screws (M12, max. length=45mm) should be precisely positioned.

3.2 OLTC installation in the bell-type transformer

Warning!	
	Please check closely whether the vacuum interrupters are broken or not!
	SHGV is a vacuum OLTC. During the installation, pay extra attention to the vacuum interrupters!
	➤ If the vacuum interrupters are broken, please contact Huaming After-sales Department immediately!

Warning!	
	Danger of injury if heavy parts are on a rough place!
	Unstably-placed tap selector might tip and cause severe injuries.
	➤ The tap selector must be put on a flat place. ➤ Avoid the tap selector tipping.

Warning!	
Heavy parts on a rough place may lead to property damage!	
Unstably-placed tap selector might tip and cause itself or other parts damaged.	
➤ The tap selector must be put on a flat place. ➤ Avoid the tap selector tipping. ➤ Use means of transport and hoists with a carrying capacity of > 500 kg.	

3.2.1 Tap-changer head flange disassembly and the switch insert lifting

1. Make sure OLTC is in the adjustment position.
2. Unscrew the 24 M10 bolts on the OLTC head cover (with spring washers, wrench 17).
3. Disassemble the OLTC head cover (See Figure4). During disassembly and other operations, avoid damaging OLTC head cover and seal surface (O shape seal ring, see Figure4).

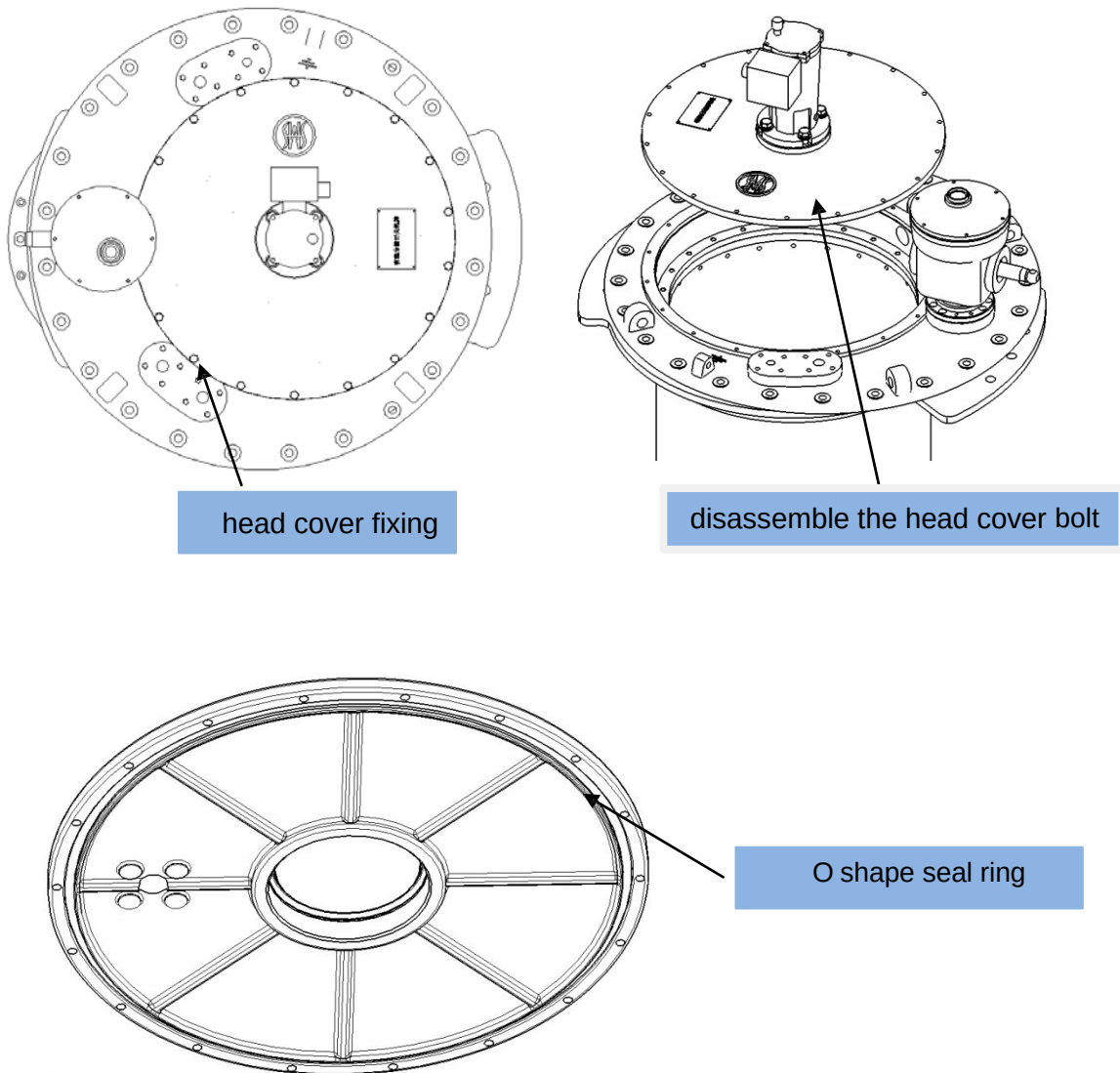


Figure 4

	Notice	Head cover seal ring must be well preserved.
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4. Remove the retaining nut (non-red area) on the flange ring (9 M8 nuts, wrench 13, with spring washers) (See Figure5).

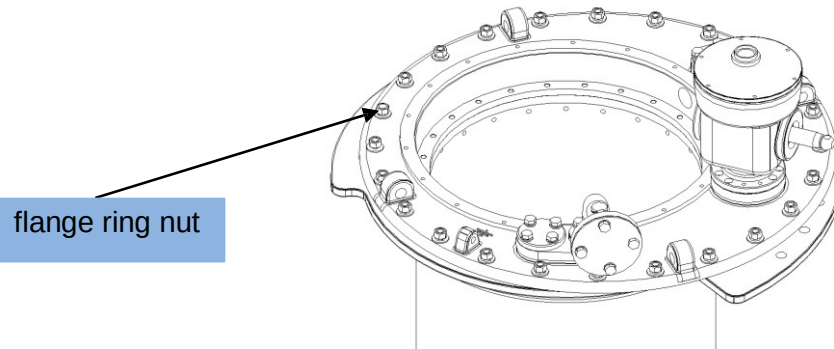
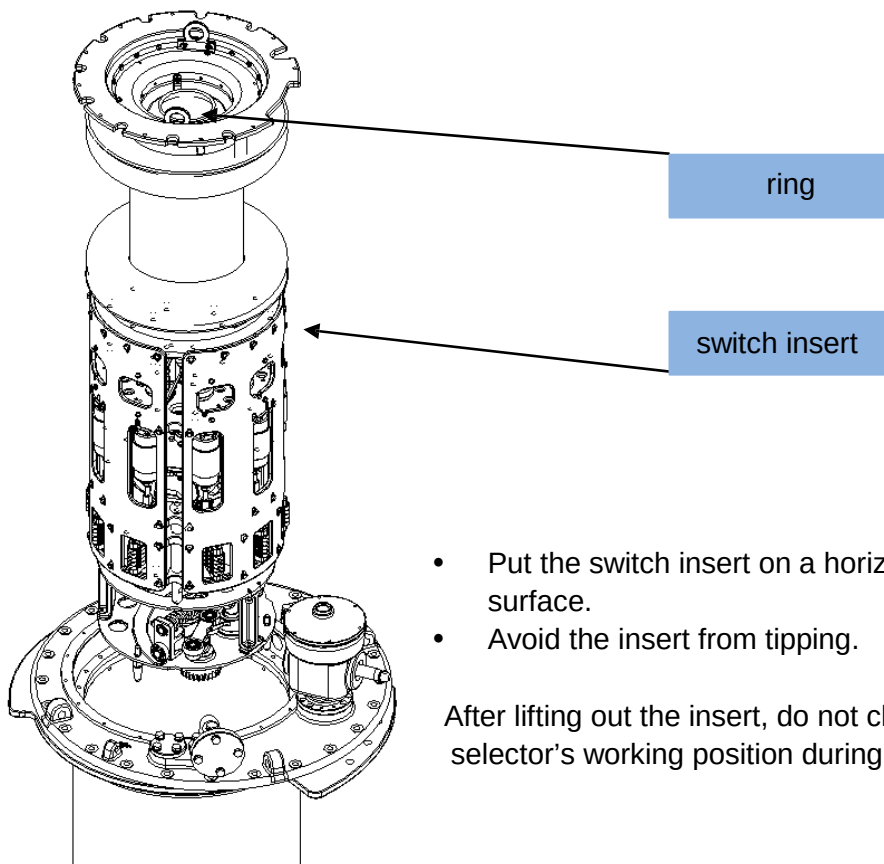


Figure 5

	Notice	Fasteners must be well preserved and avoid falling into the gas compartment.
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5. The upper part of the OLTC has a desiccant. Do not remove it. Carefully lift out the switch insert vertically from the gas compartment (See Figure 6).



- Put the switch insert on a horizontally flat surface.
- Avoid the insert from tipping.

After lifting out the insert, do not change the tap selector's working position during assembly.

Figure6

3.2.2 Lifting out installation flange

	Notice Fasteners must be well preserved and avoid falling into the gas compartment.
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Remove the remaining retaining nuts (13 M8 nuts, wrench 13, with butterfly washers) on the mounting flange (see Figure7). Lift out tap-changer mounting flange from the supporting flange (See Figure8).

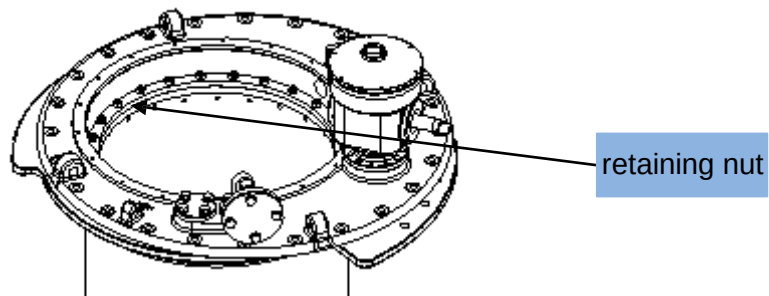


Figure7

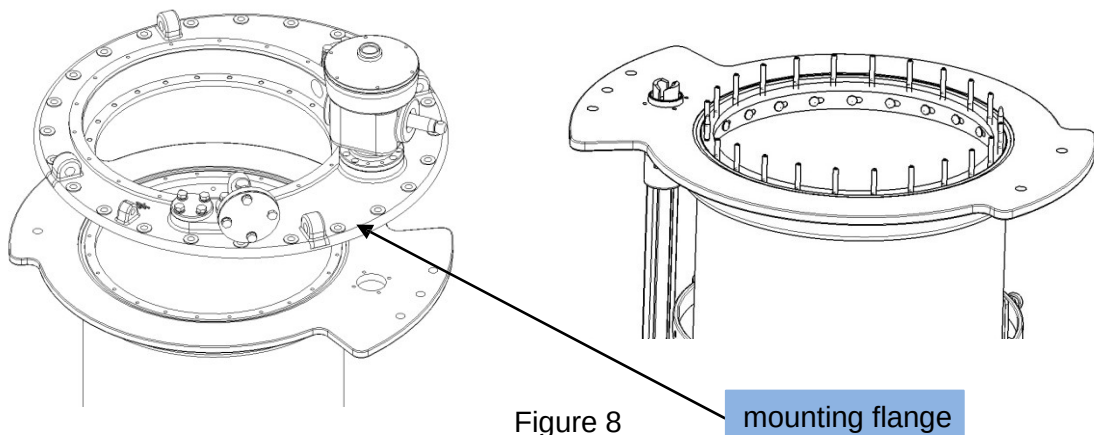


Figure 8

3.2.3 Assembly of the tap selector and diverter switch compartment

Lift the diverter switch until it's above the tap selector and be aligned with it.

1. Put the tap selector on a horizontally flat surface
2. Remove the tap selector's positioning plate. (See Figure 9)
3. Use a special tool to hoist the gas compartment of the diverter switch. (See Figure10)

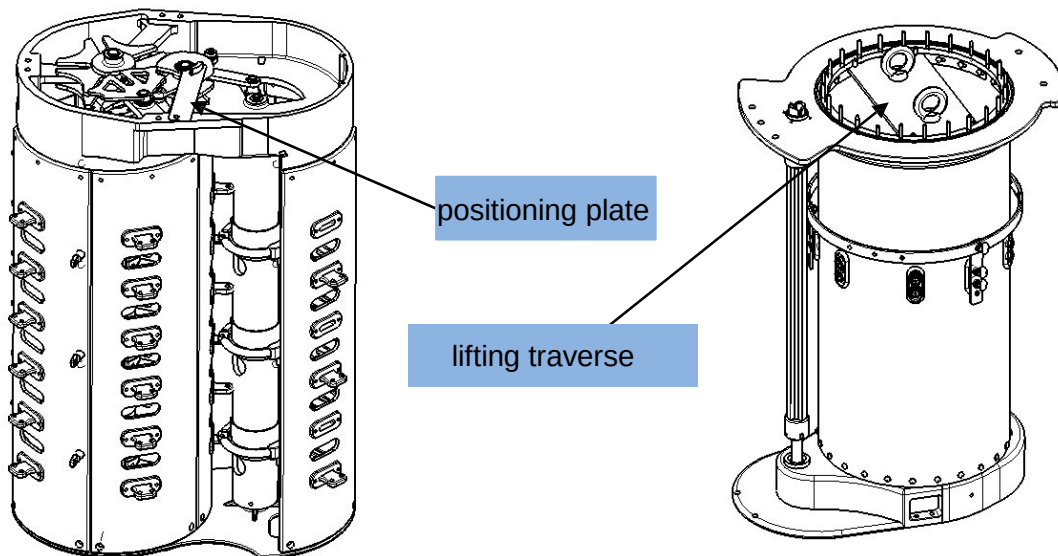


Figure 9 Tap selector's positioning plate

Figure 10 Lifting Traverse

4. Lift the diverter switch until it's above the tap selector. (See Figure11)
5. Align the two coupling parts. Stagger the upper and lower bulging parts and be careful not to bump them two.

	Notice	Be aware of the danger of incorrect alignment! After removing the positioning plates, do not move the drive wheel.
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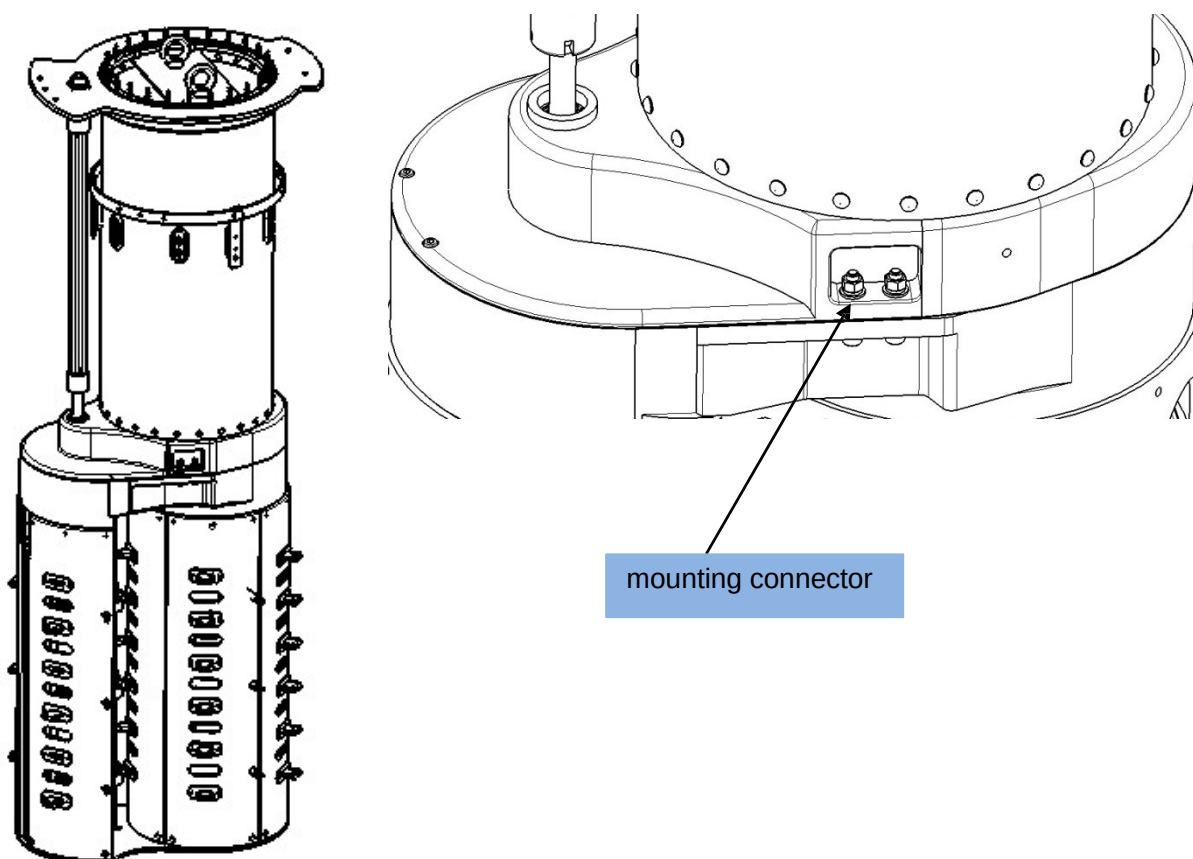


Figure11 The gas compartment and the tap selector connecting diagram

6. Align the gas compartment's mounting hole with the tap selector's. Find the mounting connector in the tap selector accessory box 1 for alignment (see Figure11) with a torque of **50 to 60 Nm**.

	Notice	The verticality of the diverter switch and the tap selector (deviation less than 2%).
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7. Connect the odd and even leads of tap selector and diverter switch gas compartment:

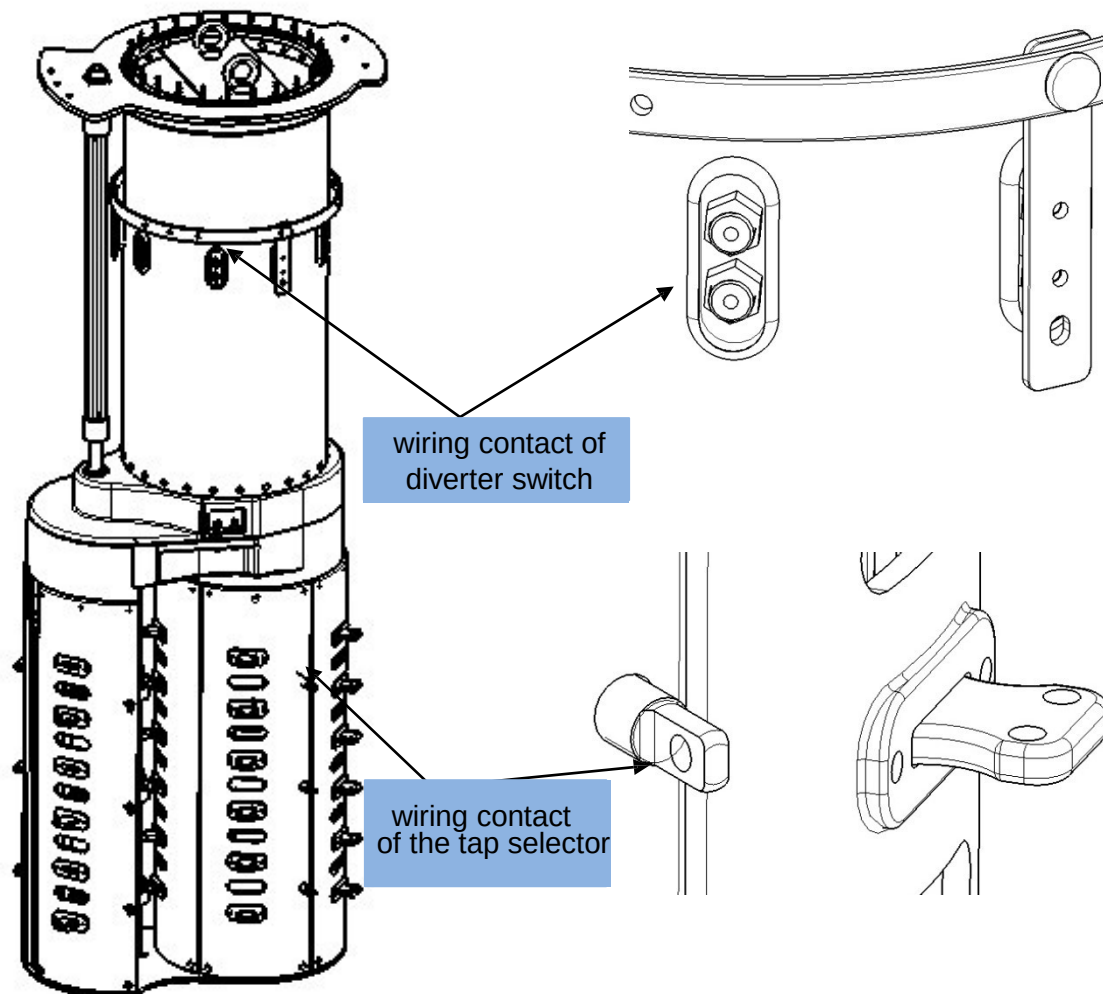


Figure 12 The connection of the even and odd leads of tap selector and diverter switch gas compartment

Warning!



Be sure to directly stick the lead wire level with contacts on insulation cylinder. Don't clip the screening cap between them (See Figure 13).



Notice

All the leads must be carefully operated. The specified tightening torque must be used to make sure the bolting is completely reliable and the terminals must be shielded with the screening caps provided together with the OLTC.

	Notice	Be careful not to damage the exterior insulation layer of leads when connecting the tap selector to the diverter switch.
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Warning!		
	When operating tap-changers, motor drive units or components, make sure they are securely fastened! Accidental tipping of the device must be avoided!	

Take out the leads in the packing box and connect the wiring contacts of diverter switch and tap selector with M8 × 25 and M10 × 35 bolts by serial number.

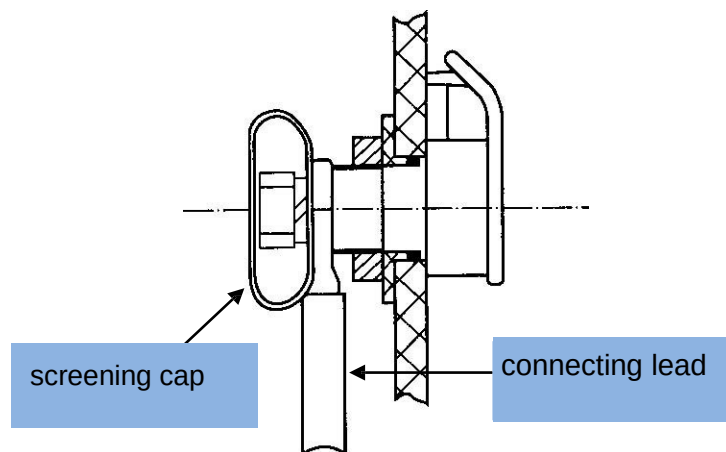


Figure 13

8. After the leads are bolted, the screening caps should be fastened.

Warning!		
	➤ Reminder! The OLTC must be in an accurate vertical position on the supporting structure. (Maximum vertical position deviation is 2°). ➤ The installation of OLTCs on the support structures. ➤ It must be ensured that the OLTC can still be raised 5-20 mm to its final position after the bell type transformer cover is closed.	

3.2.4 Install OLTC into bell type transformer tank

First, put the OLTC on the temporary holder inside of the transformer. For this, the OLTC diverter switch gas compartment is provided with a supporting flange.

Assembly steps are as follows:

1. Put the OLTC on the temporary holder and connect it to the tap winding. The OLTC should be placed on the holder stably and don't move it freely.
2. Use the mounting holes on the support flange to fasten the OLTC on the holder temporarily. If necessary, place a temporary adjustment spacer between the temporary holder and the supporting flange to adjust the installation height of the OLTC to its final installation position. The spacer must be removed before installing the bell type transformer cover.

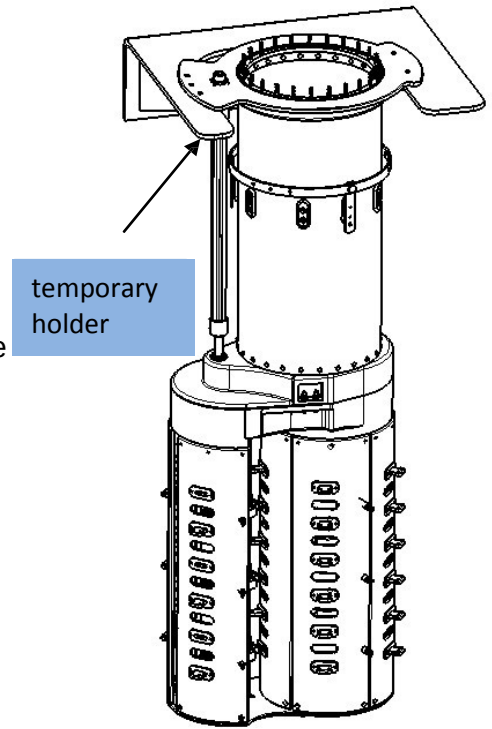


Figure 14

3.2.5 Install bell type transformer tank cover

1. Before assembling bell type transformer tank cover, clean the sealing surface of OLTC supporting flange, put a sealing ring on the supporting flange (See Figure 15), and take out the spacers.

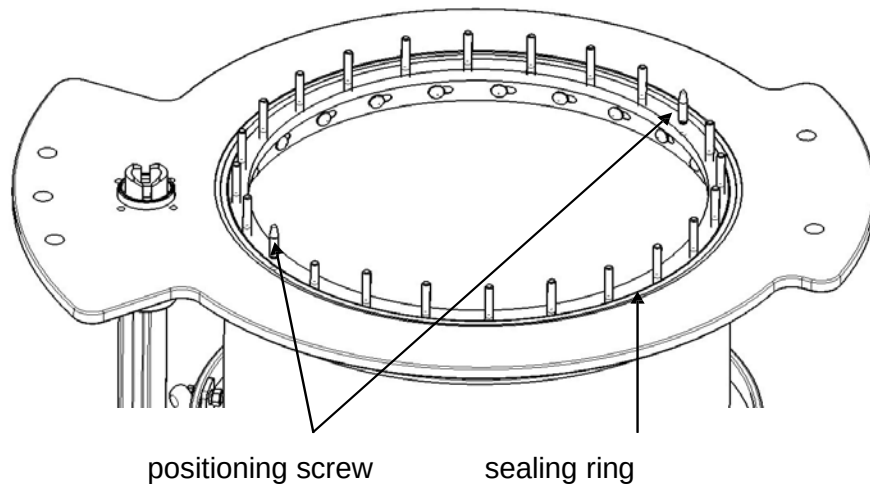
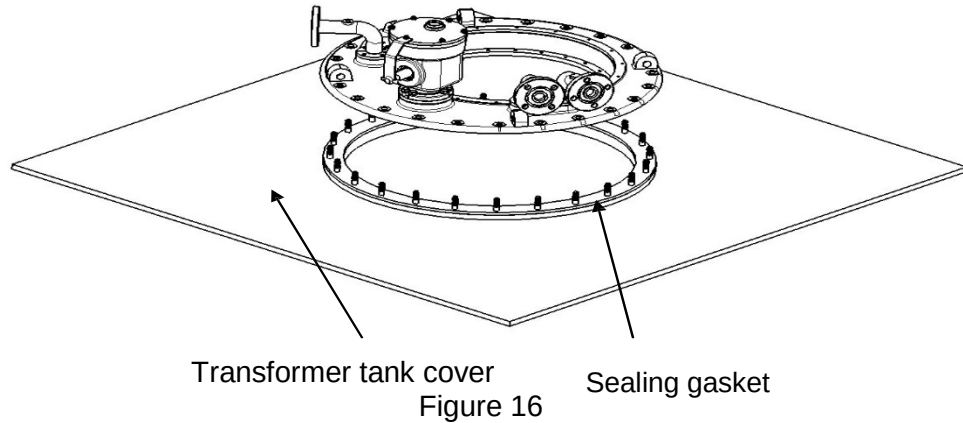


Figure.15

2. Lift the bell type transformer tank cover above the transformer active part, and assemble it.
3. Before mounting OLTC installation flange, clean the sealing surface. Place the sealing gasket on the transformer installation flange, and fix the OLTC installation flange on the transformer installation flange (24 M10 bolts, torque 100~110 N•m) (See Figure 16).



4. Slightly lift up the OLTC by special lifting device and make sure all the studs of supporting flange are accurately positioned in the OLTC fixing holes (See Figure17).
5. Install the OLTC on the head installation flange. Pay attention to the two positioning studs (with extended length), supporting flange and red triangle mark on the head cover of the OLTC. It ensures that the OLTC is installed at the correct position.

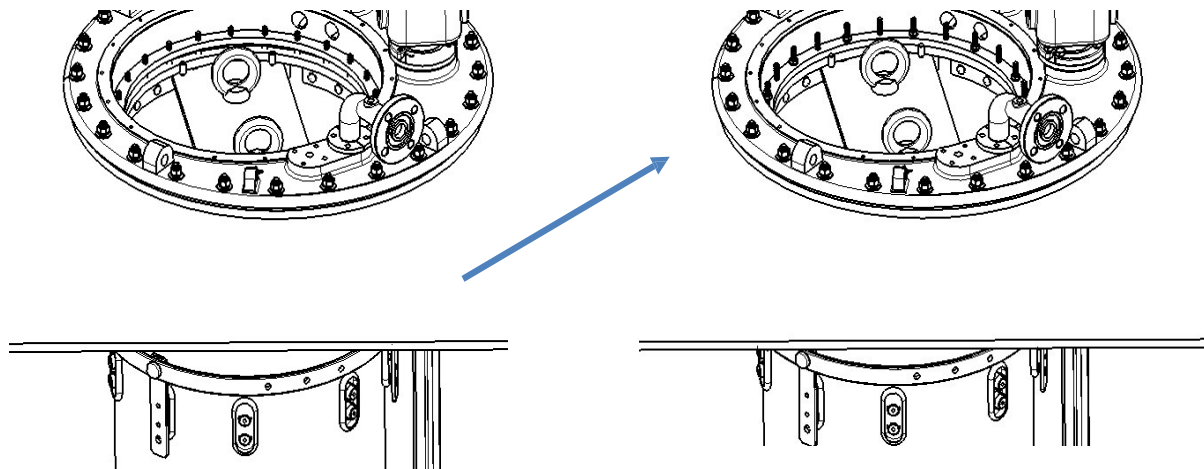


Figure 17

3.2.6 Install diverter switch insert

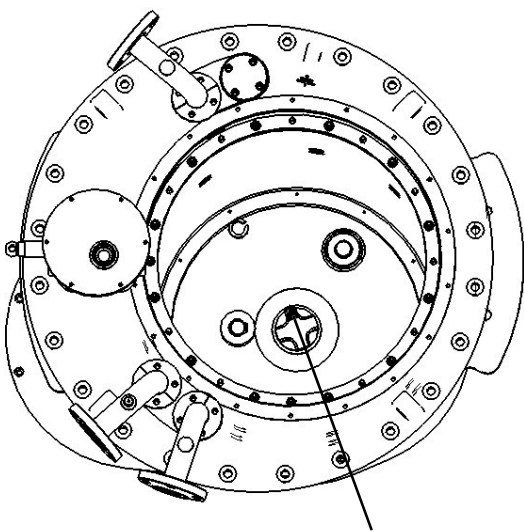
	Notice	After the installation of the insert, check the lower surface of the insert flange ring and the positioning surface of the bell flange carefully and they must be tight without gaps.
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Warning!

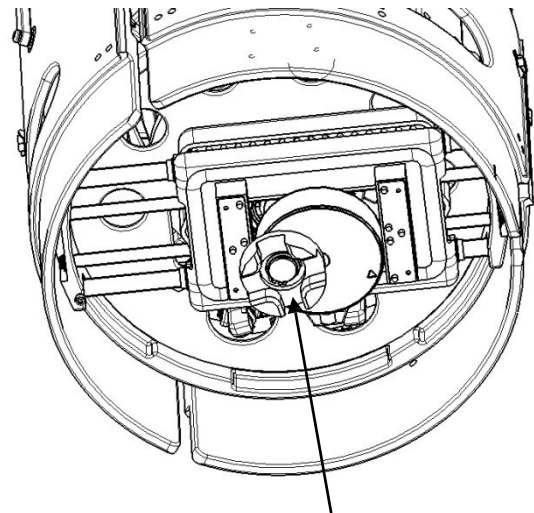


If there are gaps between the lower surface of the insert flange ring and the positioning surface of the bell flange, the OLTC may stop switching. Tap position disorders and other serious problems will also be caused!

1. Make sure that the tap selector and the bottom connectors are in the setting position when the diverter switch insert is installed, roughly aligned with the small end of the connector (See Figure 17).



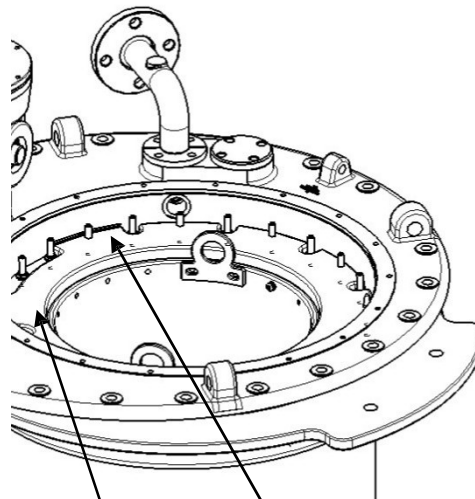
Small end of the bottom connector



Small end of the insert connector

Figure 17

2. Lift up diverter switch insert above diverter switch gas compartment and slowly put down the diverter switch insert till its final position. Fasten supporting plate by ten M8 nuts (No.13 wrench, torque 12~15Nm, with spring washer).



Flange ring

Red line

3. Installation of the head cover: put the OLTC head cover on the OLTC head flange and make sure that the sealing gasket in the cover plate is in correct position.

Notice: the sealing surface must be cleaned carefully before the installation of the head cover.

	Notice	Draw a red line on the top surface of the bell flange diverter switch insert. After the installation of the insert, the top surface of the flange ring is not allowed to exceed the red line (See Figure 18).
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3.3 OLTC installation in the standard-type transformer

3.3.1 Disassemble OLTC head installation flange and lift up diverter switch insert. →Refer to 3.2.1 for more details.

3.3.2 Lift up installation flange. →Refer to 3.2.2 for more details.

3.3.3 Assemble tap selector and diverter switch gas compartment. →Refer to 3.2.3 for more details.

3.4 Assemble installation flange and transformer tank cover

1. Before installing the OLTC installation flange, wipe the sealing surface. Place the sealing ring on the transformer installation flange and secure the OLTC installation flange to the transformer installation flange (24 M10 bolts, torque 100 to 110 Nm). (See Figure 16)
2. Lift up the transformer cover over the transformer body and assemble it.

3.4.1 Raise the SHGV by lifting device to the mounting surface of the bell flange supporting flange for connection

3.4.2 Installation methods of diverter switch insert and head cover are same as that of the bell type

3.5 Connection of tap winding leads and OLTC connection terminals

Tap winding leads must be connected according to connection diagram. OLTC connection terminal is marked with contact position code on the tap selector insulating bar. Tap winding leads and OLTC connection terminals should be in accordance with the indicating code of the OLTC.

	Notice	All the tap leads connected to the OLTC must be securely fastened. The connection of tap leads should not cause any pulling force to the connection terminals of the OLTC.
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3.5.1 Fixation of tap winding leads and tap-changer connection terminals.

Tap selector connection terminals are provided with M10 bolts through-holes, which is convenient for the connection and fixation of tap winding leads.

Screen caps supplied together with the product are intended for shielding M10 bolts (See Figure 19). Screw the nut.

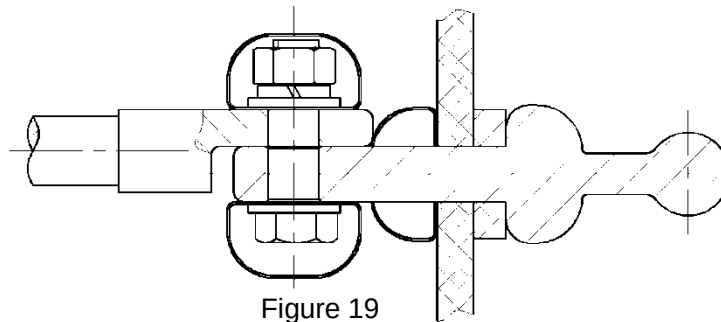


Figure 19

3.5.2 Tap selector terminal leads shall not cause forced distortion or damage to tap selector.

1. The leads of the voltage regulating winding of the transformer should be respectively led to the tap selector from both sides so as to avoid the force of the one-side lead, which will cause the deformation of the tap selector.
2. The conductor between tap selector terminal and the tap closest to the winding clamps should have some flexibility and not too short. It should be soft and not coated with insulating paint to avoid hardening of the coating after drying, which makes the insulating bar deformed under force.
3. The tap leads connected to the tap selector should be arranged in smooth circle curves, so that the insulating bar of the tap selector will not be impacted by force.
4. The connection leads of the tap selector should be lead out from the exterior of the tap selector. Never allow the wire to pass through the interior of the selector.
5. The bell type tap-changer shall be lifted up 5-20mm after the connection of the tap leads. Therefore, special attention should be given to the flexibility of connecting leads. It is recommended to install the intermediate flange on the holder, and put some spacers between the intermediate flange and the head flange, so as to obtain the required actual assembly clearance. After the lead is connected, remove the temporary spacers and check for the flexibility of the connection leads and whether the tap-changer is affected by any force.

4. Transformer ratio test and DC resistance measuring

	Notice	Connection of all leads connecting to OLTC must be firmly fastened. Assembly of connection cables must ensure that no traction would be caused by cables on the terminals.
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We would recommend carrying out a transformer ratio test and measuring DC resistance before the transformer is dried.

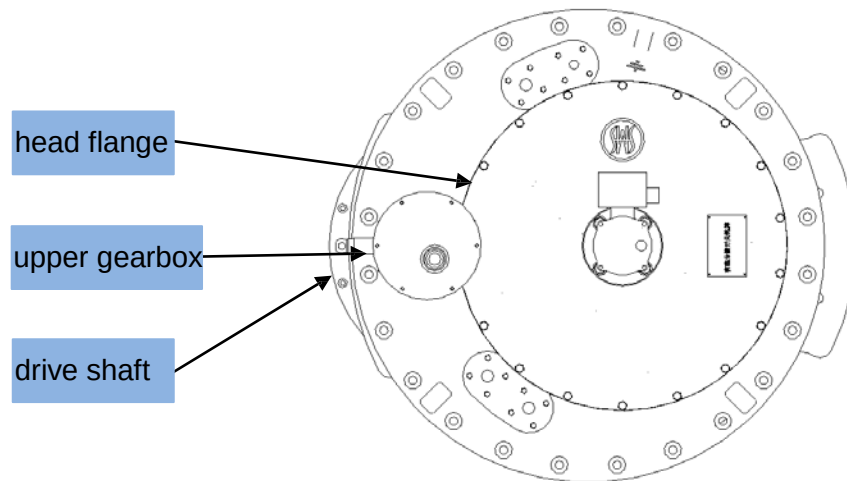


Figure 20

4.1 Transformer ratio test (mount temporarily head flange & diverter switch insert for bell type installation)

Use one short tube with inner diameter of 25mm (one connecting nut with diameter of 12mm is already screwed in) in company with a hand wheel or a crank to rotate the drive shaft of upper gearbox to realize a position change. Noise announcing the operation can be clearly heard. After having heard the noise, rotate the shaft for 2.5 turns more in the same direction to complete one full position change.

In case of synchronized operation of 2 or 3 OLTCs, it is necessary to connect all these OLTC heads by means of horizontal shaft.

When operating the OLTC manually, check operating positions through the observation window at head of OLTC in case the end position is surpassed.

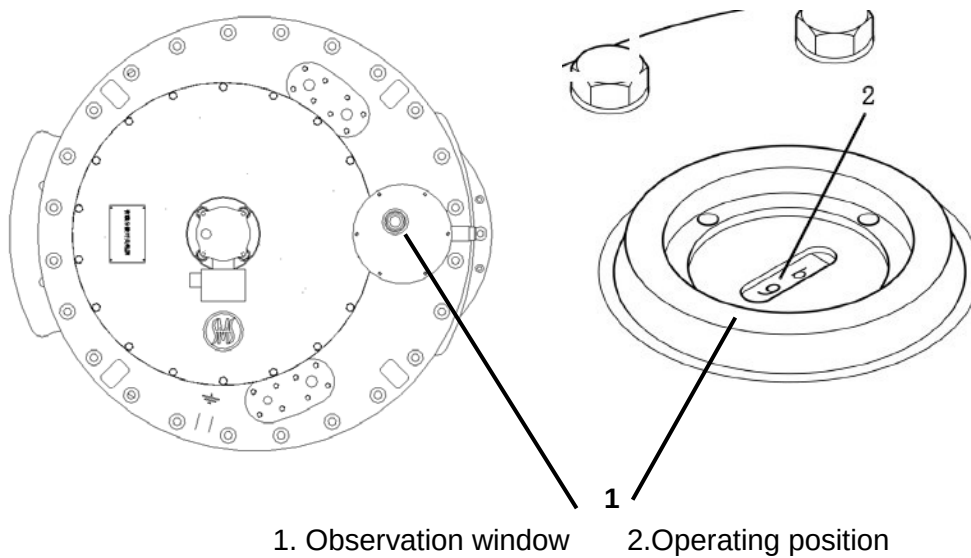


Figure 21

	Notice	Operate the OLTC only by rotating the drive shaft of upper gearbox
---	--------	--

4.2 Measuring DC resistance on transformer

	Notice	After performing the transformer ratio test and measuring the DC resistance, it is a must to adjust the OLTC to its set position.
---	--------	---

5. Drying process

	Notice	1. Prevent small parts from falling in the diverter switch oil compartment. 2. Check you have the same number of small parts when disassembling and reassembling. 3. Check that the sealing surfaces on the OLTC head cover and OLTC head are not damaged during removal and during all
---	--------	---

To guarantee the dielectric properties of OLTC, a minimum drying process must be carried out as described below (alternatively vacuum-drying or vapor-phase drying).

	Notice	1. Pay attention to metal particle which must be removed. 2. With head cover removed, avoid foreign particle coming inside the oil compartment. 3. Pay attention to the eventual temperature and ensure that the temperature increase speed should be controlled acc.to the value stipulated hereinafter. 4. Temperature measuring point should be properly closer to the OLTC. 5. After the drying, isolate the OLTC from atmosphere for 48 hours to avoid influence of humidity in the air.
---	--------	---

5.1 Vacuum drying

Remove OLTC head cover including upper gear unit, motor-drive unit, protective relay, pressure-operated relay, pressure relief valve, bevel gear, temperature sensor, oil filter unit before drying and keep outside autoclave.

5.1.1 Vacuum drying in autoclave

1. Heat up the on-load tap-changer in air at atmospheric pressure with a temperature rise of approx. 10 °C/hour to a maximum final temperature of 110 °C.
2. Pre-dry the on-load tap-changer in circulating air at a max. temperature of 110 °C for a period of 20 hours.
3. Vacuum dry on-load tap-changer at between 105 °C and maximum 125 °C for at least 50 hours. Residual pressure should be no more than 133Pa.

5.1.2 Vacuum drying in transformer tank

1. Establish a connecting lead (rated diameter: 25 mm) between the connections E2 and Q on the on-load tap-changer head.
2. Heat up the on-load tap-changer in air at atmospheric pressure with a temperature rise of approx. 10 °C/hour to a maximum final temperature of 110 °C.
3. Pre-dry the on-load tap-changer in circulating air at a max. temperature of 110 °C for a period of 20 hours.
4. Vacuum dry on-load tap-changer at between 105 °C and maximum 110 °C for at least 50 hours. Residual pressure of no more than 133 bar.

5.1.3 Treatment on OLTC after drying process

1. After drying process, it is necessary to verify if there is foreign particle on surface of the contacts of tap selector. Meanwhile, ensure that the lubricant is still on the surface of contacts during drying.
2. After drying process with transformer, it is permitted to add a bit lubricant on motor drive unit and active part of contacts to prevent dry abrasion.

6. Installing protective device and drive mechanism accessories

6.1 Installation of rapid pressure rise sensor

Please follow the specific operation instructions of rapid pressure rise sensor offered by the supplier.

6.2 Installation of motor drive unit

	Notice	The mounting plate for motor drive unit must be flat and proper. Otherwise the motor drive unit can be deformed, which undermines the use.
---	--------	--

The motor drive unit is applied to drive operation of the OLTC in either manual or automatic mode Please pay attention to the following points when fitting the OLTC with motor drive unit:

It is a must to connect the OLTC with motor drive unit at the set position which is indicated on the connection diagram supplied together with the equipment.

The motor drive unit is mounted vertically on transformer tank, which permits no skewness.

Fit motor drive unit as described in relevant HM operating instructions of type SHM or CMA7.

6.3 Installation of bevel gearbox

Fix the bevel gearbox on the holder connected to the transformer cover by means of 2 bolts (aperture: 18mm) .

	Notice	The stamped serial number of the bevel gear must match the serial number of the on-load tap-changer.
---	--------	--

It is necessary to install middle support gearbox if either horizontal or vertical shaft is longer than 2m. The information provided above applies analogously to special design bevel gears and angle gears, as well as to intermediate bearings of the vertical or horizontal drive shaft.

Fix the bevel gearbox on the mounting flange with 3 pressing plates (each of them is coupled with 2 screws). It is possible to adjust its angle following user's demand:

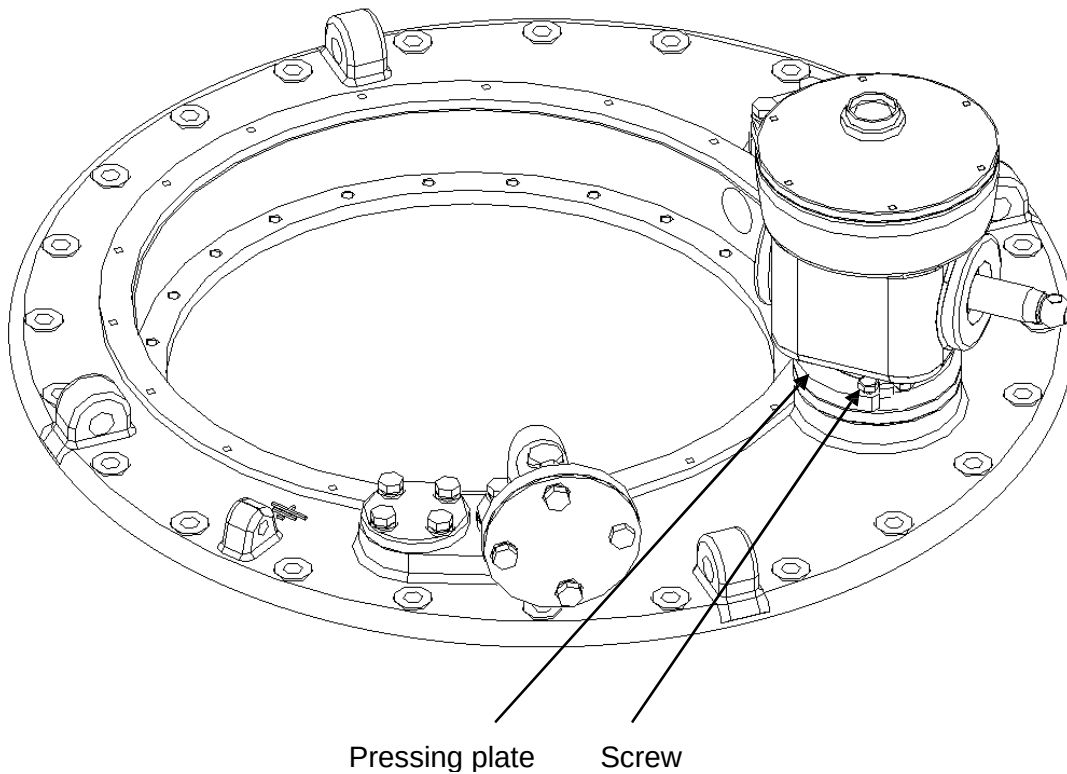


Figure22

1. Loosen the pressing plates and move out one end of the plates a bit (6 pcs. M8 screws by #13 spanner).
2. Revolve horizontally the gearbox to the required azimuth through the output shaft of the gearbox.
3. Tighten the pressing plates and fasten the screws (torque: 15 Nm).

6.4 Installation of horizontal and vertical shafts

1. Install the horizontal and vertical shafts according to the attached instructions

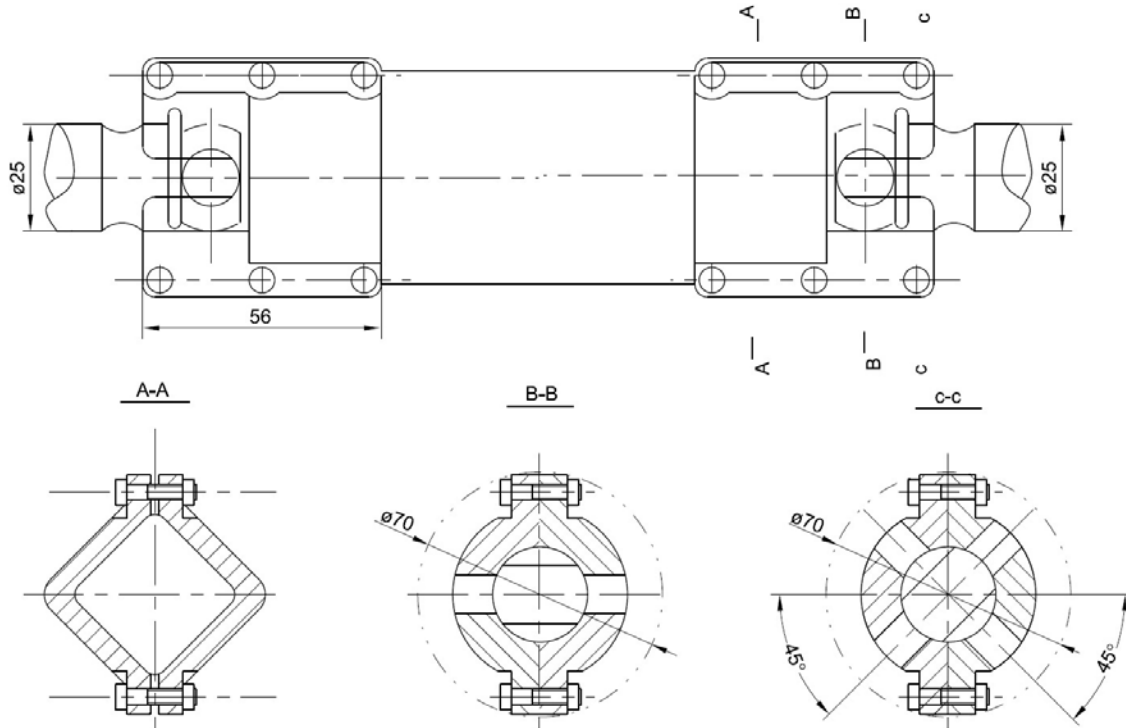


Figure23

2. Connect the motor drive unit and OLTC with the shafts according to the operation instructions of the motor drive unit (Figure 23).
3. If the vertical shaft is longer than 2m, it is necessary to install middle support gearbox (Figure 24) .

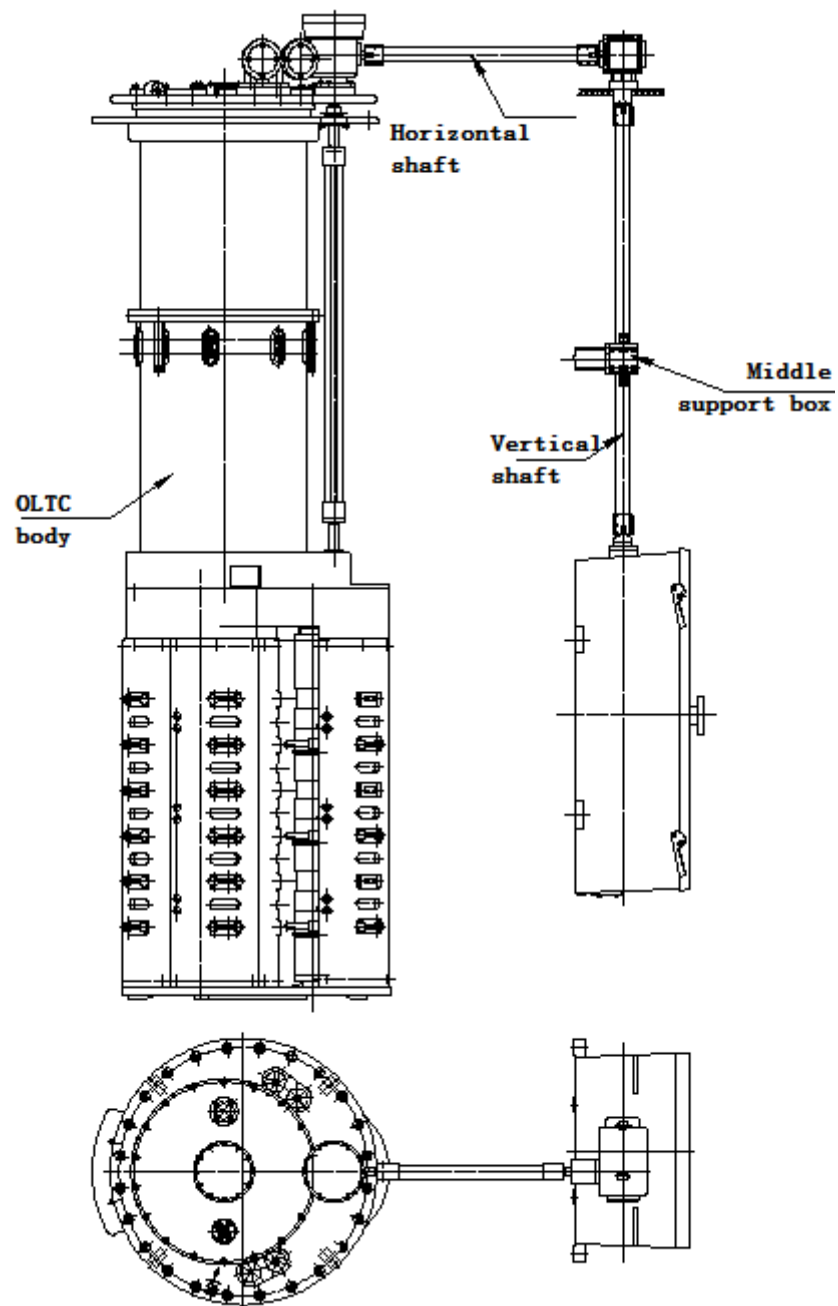


Figure24

6.5 Installation of protective cover

6.5.1 Installation of horizontal protective cover

The length of horizontal protective cover is designed and decided by transformer factory. After installed drive shaft, put clamping band on two ends of horizontal protective cover (will be provided), and clamp it on the bumping pieces of bevel gear box and gear box. Move and tighten the clamping band to the bumping pieces (Figure 25)

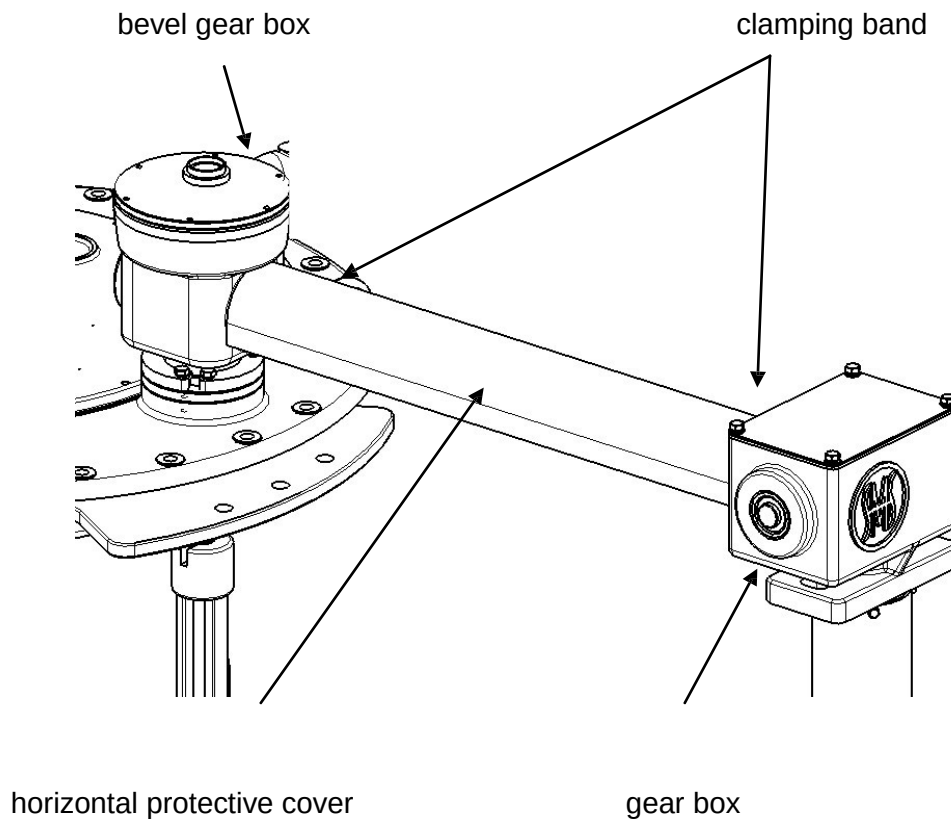


Figure 25

6.5.2 Installation of vertical protective cover (Figure 26)

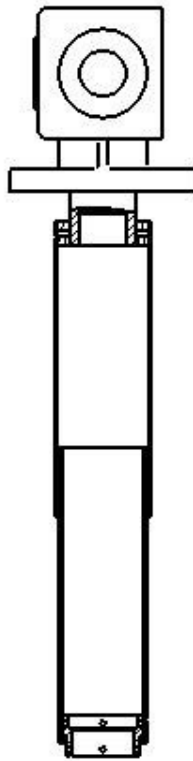


Figure 26

6.6 Adjustment position of OLTC drive system

Having connected the tap changer with motor drive unit, the mechanism should first be manually operate a full cycle of operation taps.

When the tap changer has been connected to the motor drive unit, the time elapsed between the completion of selector switch and the ending of the motor drive unit operation should be the same in both direction of rotation.

Generally, the verification of the connection of the tap changer to the motor drive unit has been down in the factory. However, for proper operation of the tap changer, the verification shall still be performed by the client before operation.

The verification of connection is carried out according to the following procedure:

1. Rotate the handle in the 1→N direction. After the diverter switch has operated (after you hear switching in oil compartment), turn the handle continuously and record the number of turns until a red mark reaches the green area on the MDU position indication window. Take the number of turns as m.
2. Turn the handle in the reverse direction N→1 to return to its setting position. Record the number of turns K in the same way as mentioned above.

3. The connection will be correct if $m=K$. If $m \neq K$ and $m-K > 1$, then the difference of turns shall be compensated. Loosen the vertical driving shaft, turn the handle $1/2(m-K)$ turns in the direction of increment of turns, and finally connect the vertical driving shaft to the motor drive unit.
4. Check the difference of turns between the motor drive unit and the tap changer in the same way as mentioned above, until the same number of turns, i.e. $m=K$ reaches.

Example:

- The verification of the connection of CM tap changer and CMA7 motor drive unit: Turn from position 10 (setting position) to position 11, $m=5$ turns, Turn backward from position 11 to position 10 (the original setting position), the $K=3$ turns. The difference of turns of the handle $m-K=5-3=2$ turns.
- Turns to be adjusted $1/2(m-K)=1/2(5-3)=1$ turn.
- Loosen the connection between the vertical driving shaft and the motor drive unit. Turn the handle in the direction 10→11 for one turn.
- Reconnect the driving shaft.
- Check that the difference of turns in both directions has been balanced.
 - a. Record number of turns m and K under connected condition.
 - b. Turn $1/2(m-K)$ turns in the direction in the increment of turns during loosening of connection.
 - c. Again make connection and verify until $m=K$.

7. SF6 gas filling

Follow the below procedure to fill the SF6 gas into diverter switch compartment.

- 1) Ensure the valve between break out pressure replay and OLTC is open.
- 2) Connect the vacuum pump and the inflation and deflation valve of diverter switch, and make the vacuum condition in the gas compartment. The gas compartment should keep in vacuum condition for at least 30 min after the vacuum pressure reach 600Pa.

	Notice	The pressure differentiation between diverter switch and transformer should not exceed 0.3MPa in any condition (gage pressure).
---	--------	---

- 3) Filling SF6 gas through the inflation and deflation valve of gas compartment in diverter switch. The amount of gas should follow the below figure.

	Notice	When new adsorbent or just change the adsorbent applied, the inflation amount should follow the figure 27 add 0.005MPa
---	--------	--

- 4) Ensure there is no leakage on every flange and OLTC head cover through leak detector after gas inflation.

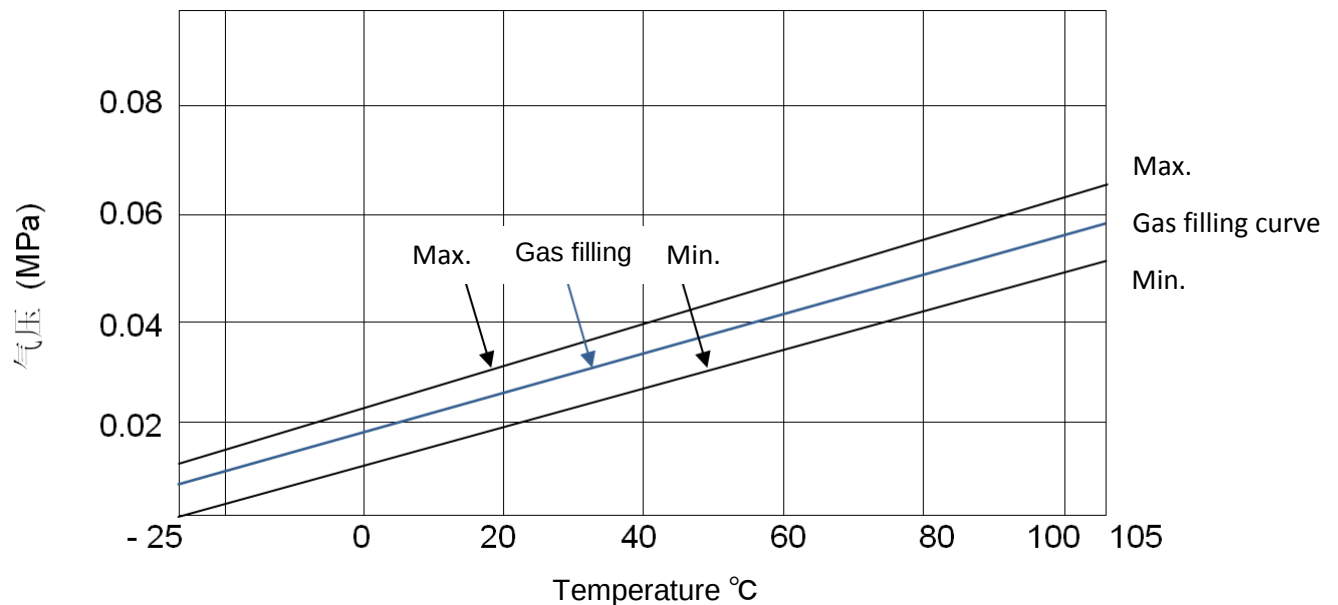


Figure 27 Gas Filling Curve

8. OTLC type test and preparation before test

8.1 Preparation before test

8.1.1 OLTC pressure inspection

8.1.2 Grounding connections

1. Connect the grounding screws of the OLTC head (1 screw M12 with nuts, wrench 19, torque 50-60Nm) to the transformer cover (Figure 28).

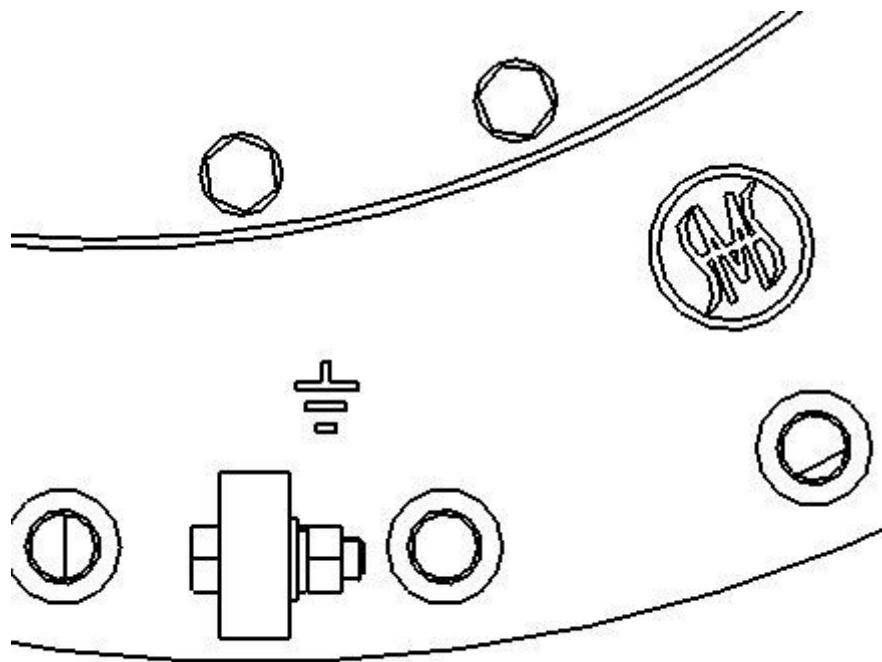


Figure 28

2. Connect the grounding screw (M12, wrench 19, torque 50-60Nm) of the motor drive cabinet to the transformer tank.

8.2 Test in transformer factory

8.2.1 Operation test

	Notice	Before transformer is energized, a test operation must be performed to ensure the mechanism function of OLTC and MDU.
---	--------	---

		Notice
	Incorrect or improper position between tap changer and motor drive unit, continue operation will lead damage of tap changer!	
	Must make sure that motor drive unit and tap changer point at same position on position indicator!	

1. During operation test, tap changer must go through whole tap range.
2. Make sure MDU and OLTC (position indicator) are point to same tap position.
3. Test the mechanical and electrical limit switch at both Max. and Min. position (Please see detail in MDU operation instruction)

8.2.2 Electric test of transformer

Warning!	
	There is explosive gas at OLTC break out pressure relay might cause the danger of fatal and serious injury.
Please ensure:	
➤	
There is no flame, heat source or sparks (e.g., through static charging) at surrounding.	
➤	
Make sure that tap-changer test must be carried out by professional test staff according to test guidance.	

9. Inspection before operation

The following inspections should be applied before transformer operation, or before the re-operation of transformer which haven't worked for a long time.

9.1 Visual inspection

- Check whether the gas density indicator of OLTC is 0.025MPa.
- Check whether the position showed in OLTC head indicator match with MDU position indicator, must be adjust if not.
- Check whether the bolts of exterior connection are tightened.
- Check whether the wire of MDU is correctly connected.
- Check whether the ground is well connected.

9.2 Inspection of manual operation

- Use hand crank to operate the MDU, check whether normal sounds appear or not.
- Must use the hand crank to check the limit operation, whether the OLTC operate normally.
- After 33 circles operations, MDU should indicate in the grey zone.

9.3 Inspection of electric operation

Close the air switch, then push up or down, make sure MDU operate correctly, and all the control device are operate accordingly to requirement. MDU should in the set position after the electric operation.

9.4 Transformer ratio measurement test

Transformer company must apply the transformer ratio measurement test before operation, and the test result should be match with HM's type test.

10. OLTC operation monitoring and maintenance

10.1 Operation monitoring

In order to make sure OLTC can operate normally, it is necessary inspect the exterior condition of OLTC with transformer. The following inspections should be applied:

- The pressure of compartment whether it is normal
- The sealing condition of MDU whether in good condition
- The heating device of MDU whether in good condition
- The sound of OLTC switch whether it is normal
- The counter movement of MDU whether it is normal

10.2 The maintenance of OLTC

Warning!		
	The maintenance should be apply by trained professional with qualification from OLTC manufacturer.	
	Incorporate maintenance will lead damage of OTLC or injury of staff.	

	Notice	Make sure that the power of transformer is cut off and grounding well connected before maintenance
---	--------	--

	Notice	OLTC should maintain periodically, to ensure it perform well. Otherwise OLTC and transformer might damage.
---	--------	--

10.3 Routing inspection

Checking the density meter, and the diverter switch whether it sound normal.

10.4 Periodic inspection

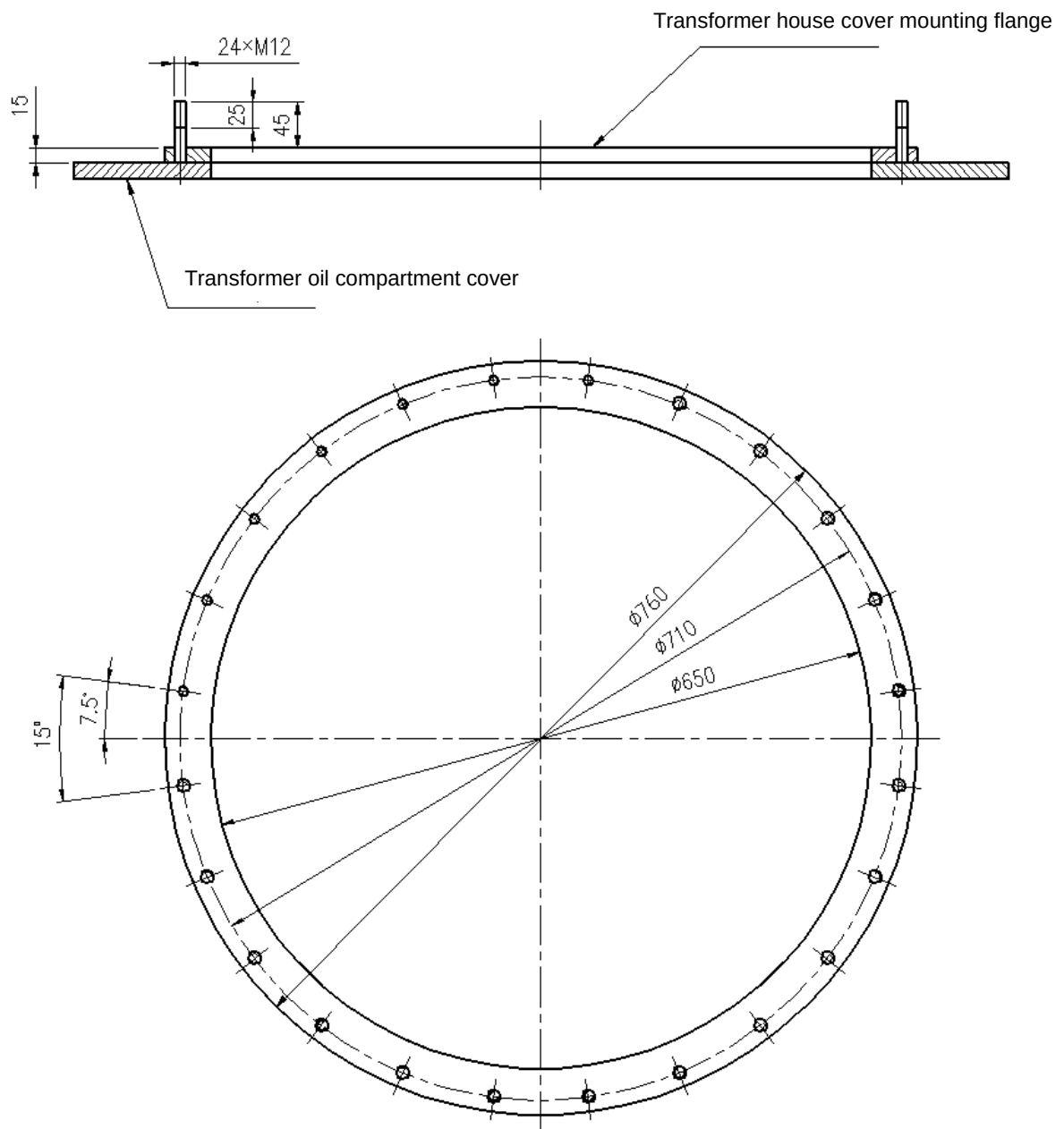
- Inspect the OLTC after 20,000 operations or after operation 2 years which comes first.
- The diverter switch insert must be replace after 300,000 operations.
- The OLTC should be applying the inspection after the 800,000 operations of tap selector.

The inspection should apply by Huaming and usually take one day to accomplish.

Appendix

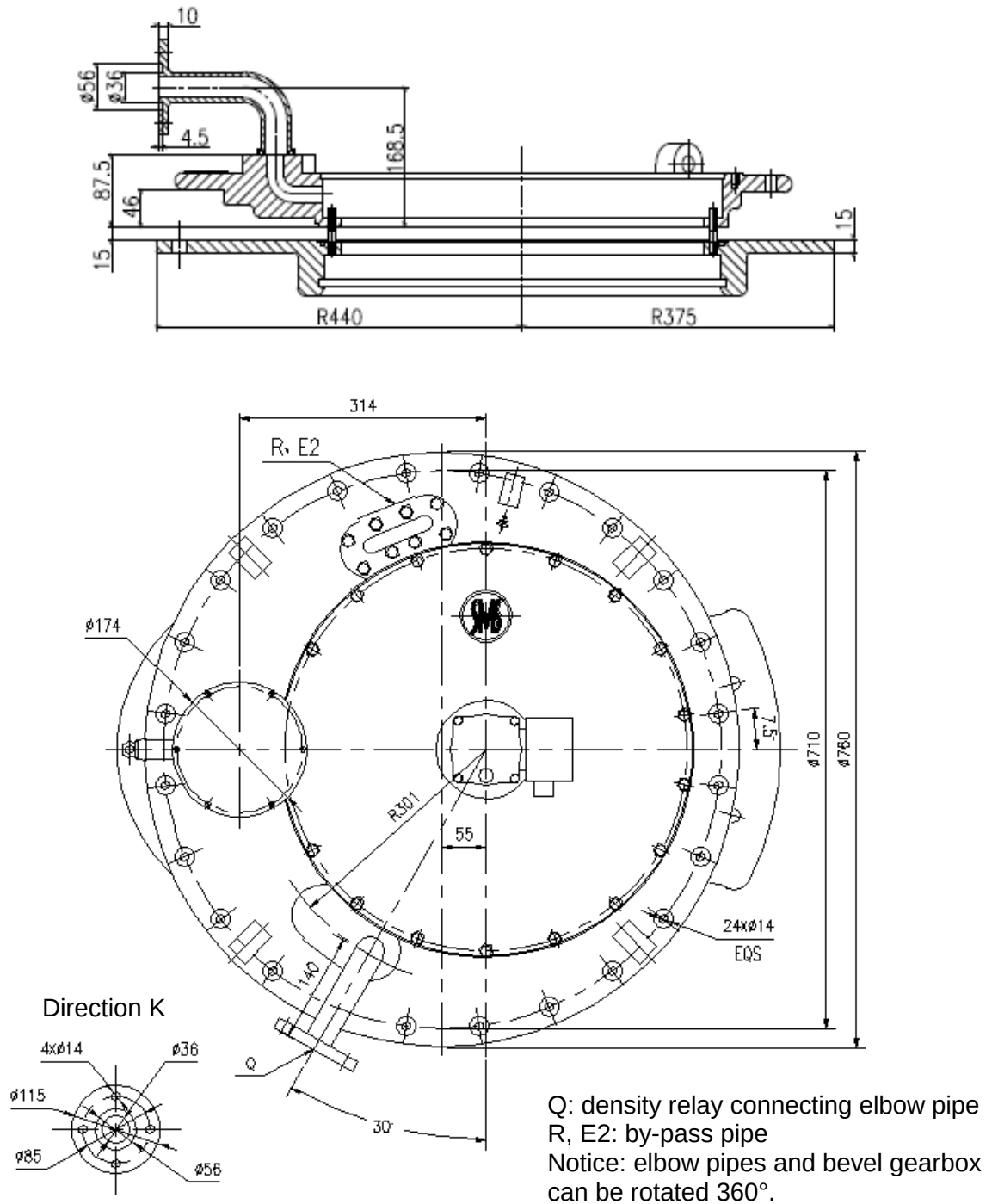
Appendix 1 SHGV OLTC transformer head cover installation flange, overall dimensions.....	42
Appendix 2 SHGV OLTC head flange installation for standard type, overall dimensions.....	43
Appendix 3 SHGV OLTC head flange installation for bell type, overall dimensions.....	44
Appendix 4 SHGV OLTC overall dimensions of SHGVIII-400Y/126C-10193G.....	45
Appendix 5 Transformer connection flange for SHGV OLTC, overall dimension....	46
Appendix 6 4:1 Gear box dimension	47
Appendix 7 SHGV and SHM-III MDU connection diagram.....	48
Appendix 8 Diagram of the installation of horizontal and vertical drive.....	49
Appendix 9 Dimension of lifting plate for bell type	49
Appendix 10 Structure diagram of bypass pipe	50
Appendix 11 SHGV (10191G) operating position table and connection diagram.....	51
Appendix 12 SHGV (10193W) operating position table and connection diagram.....	52
Appendix 13 SHGV (10193G) operating position table and connection diagram.....	53
Appendix 14 SHGV (10193G) operating position table and connection diagram.....	54

Appendix 1 SHGV OLTC transformer head cover installation flange, overall dimensions

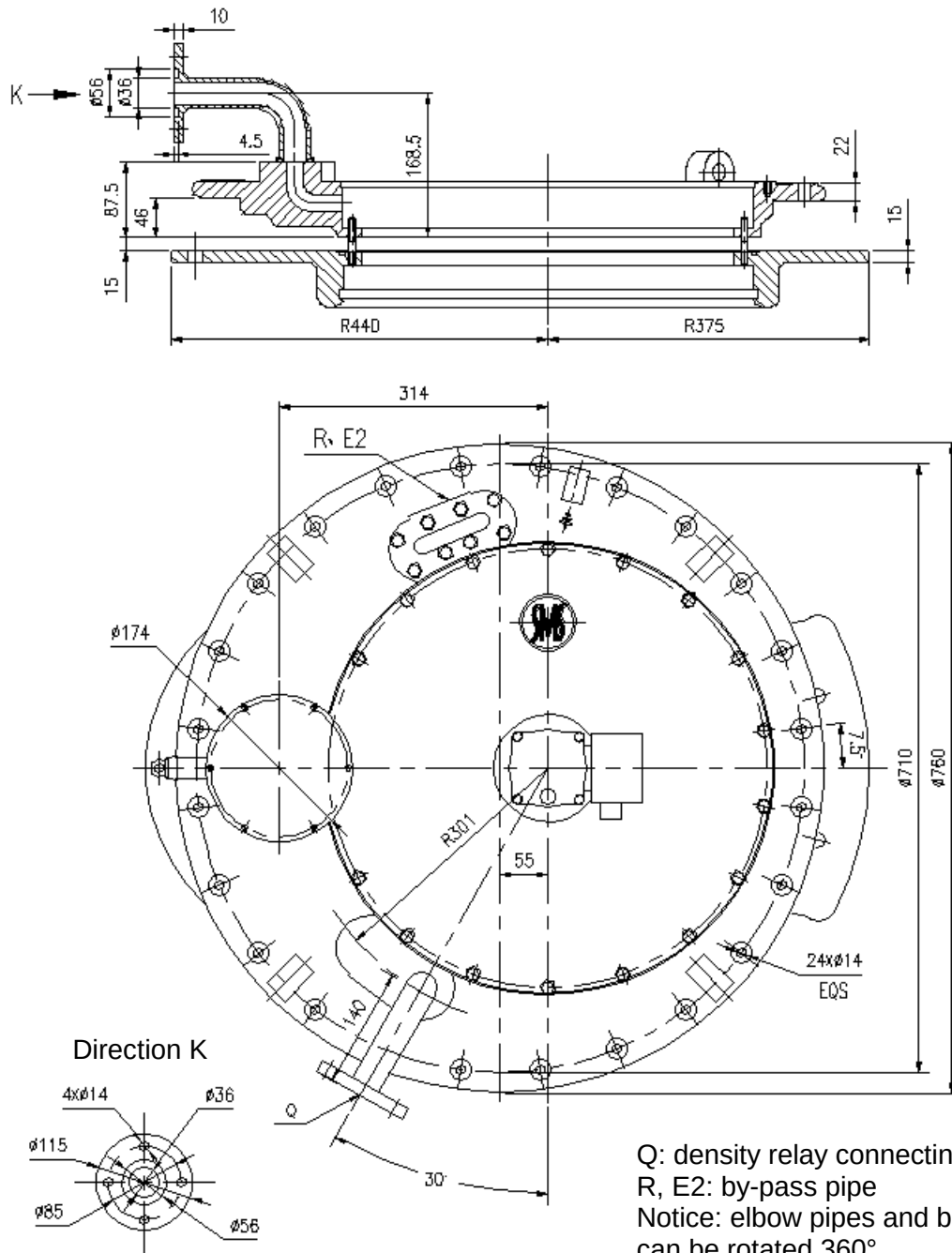


Appendix 2 SHGV OLTC head flange installation for standard type, overall dimensions

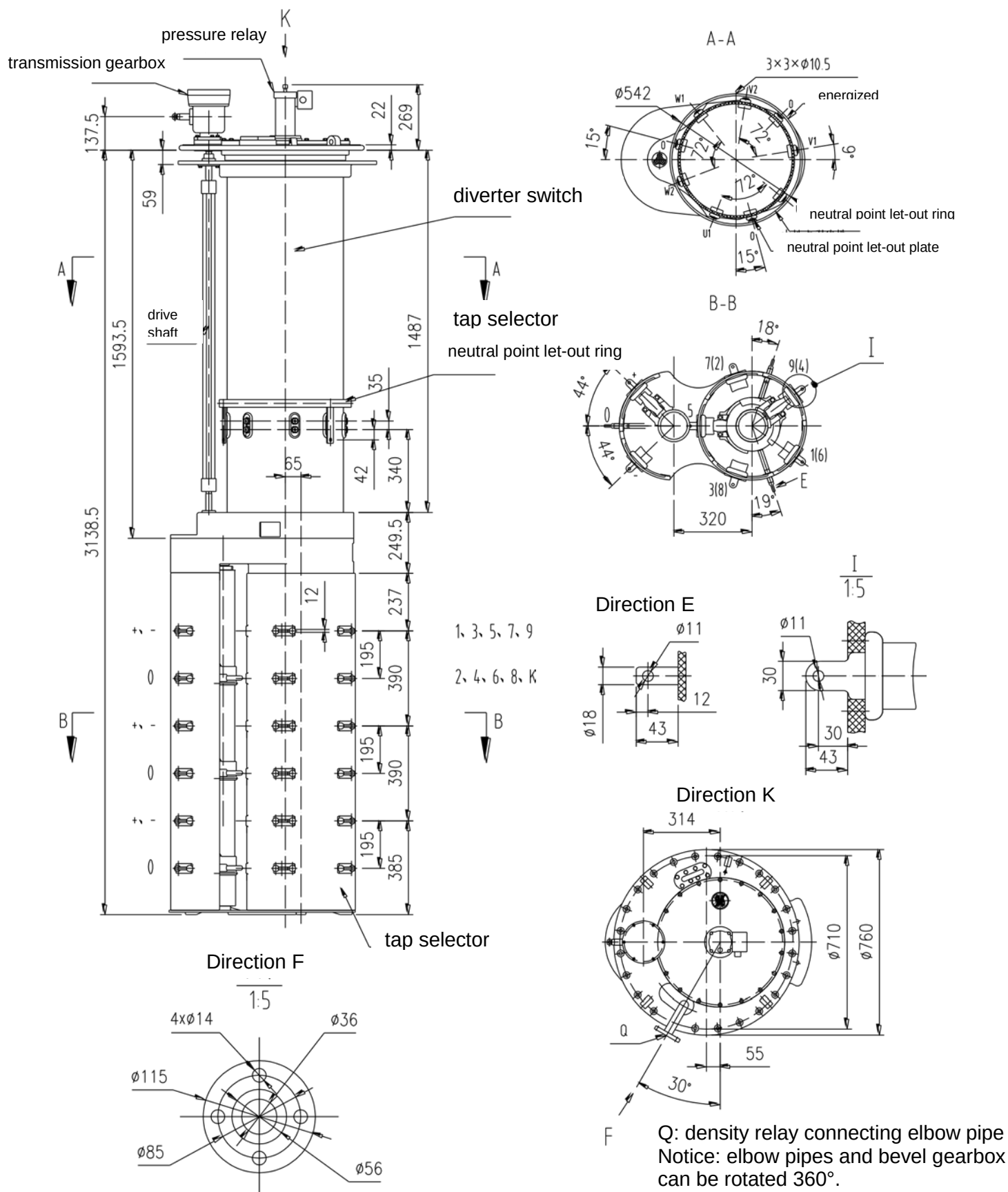
3



Appendix 3 SHGV OLTC head flange installation for bell type, overall dimensions



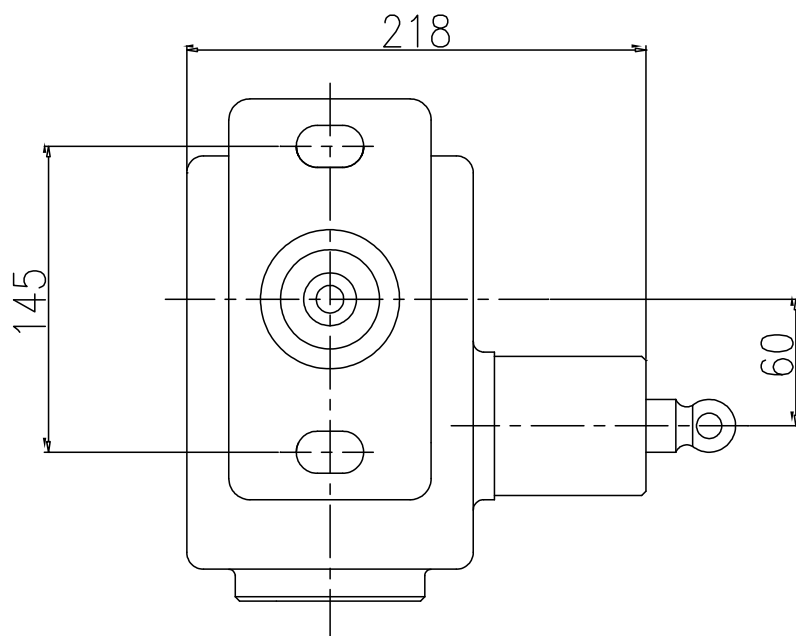
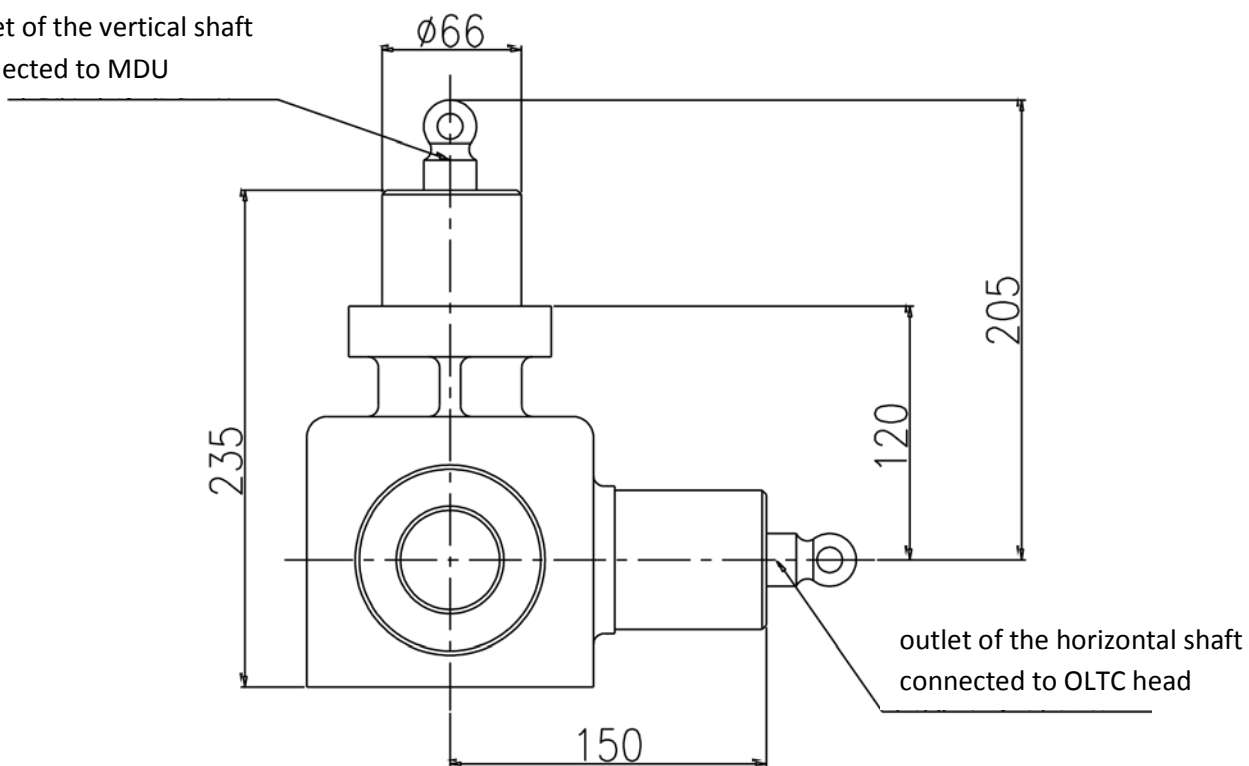
Appendix 4 SHGV OLTC overall dimensions of SHGVIII-400Y/126C-10193G



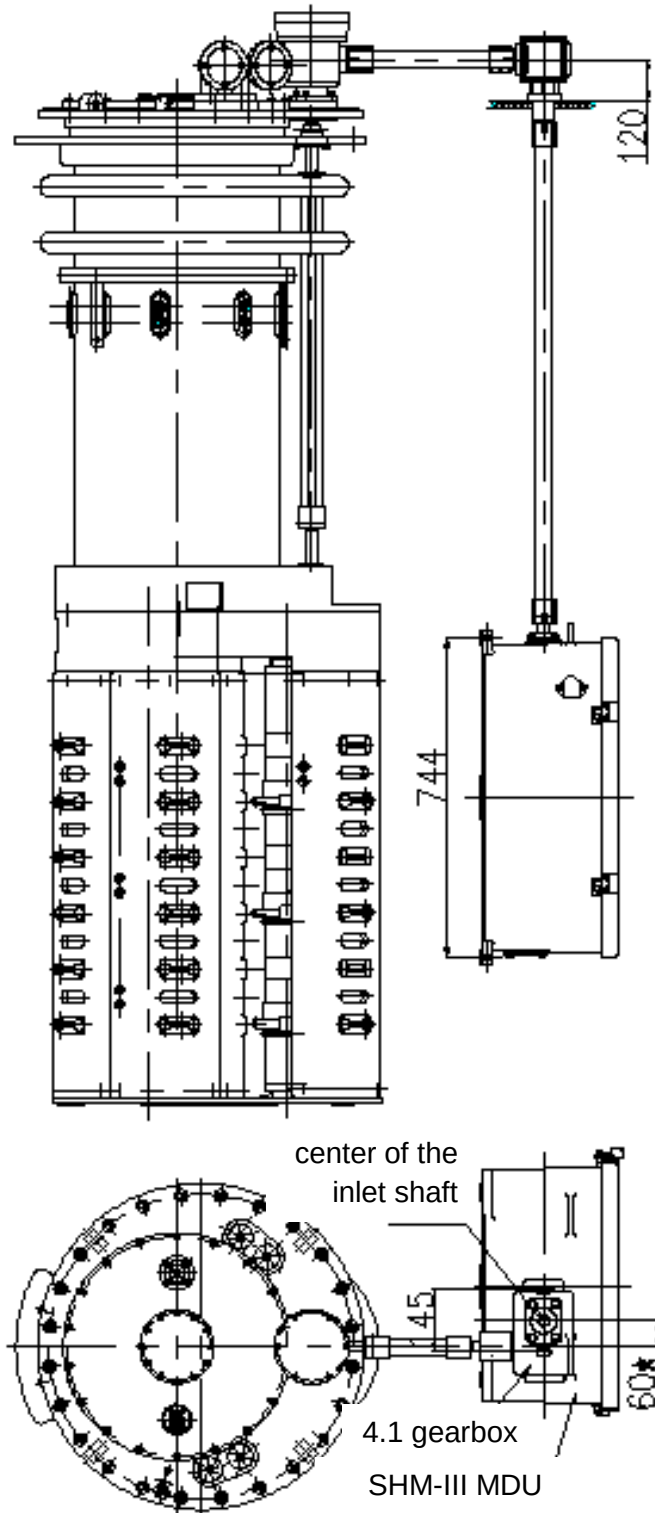
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Appendix 6 4:1 Gear box dimension

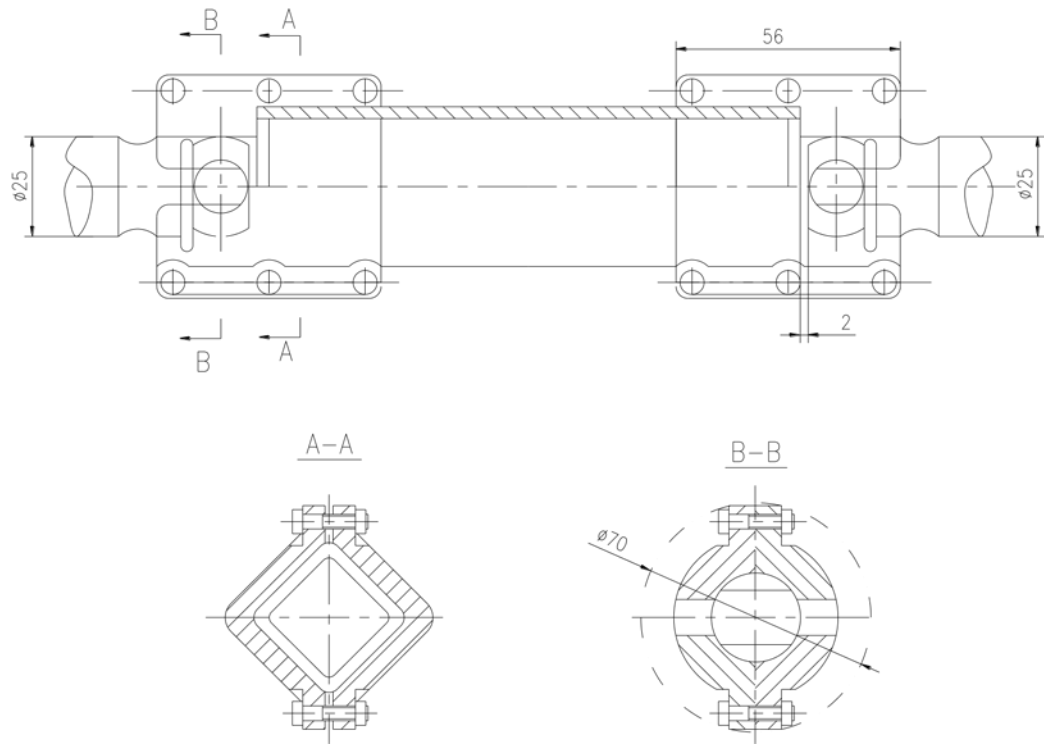
outlet of the vertical shaft
connected to MDU



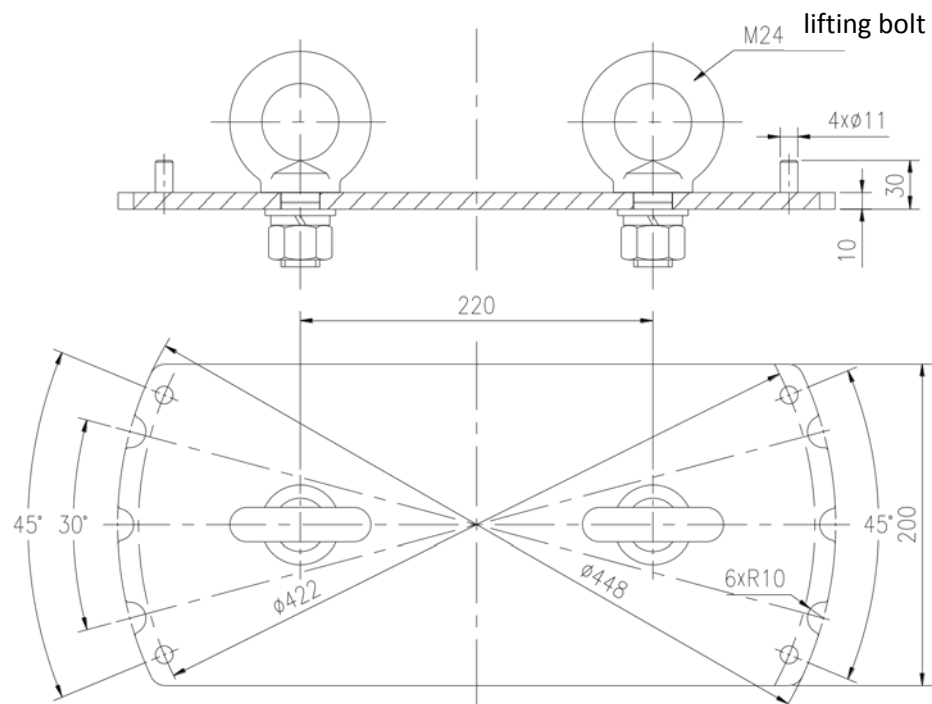
Appendix 7 SHGV and SHM-III MDU connection diagram



Appendix 8 Diagram of the installation of horizontal and vertical drive

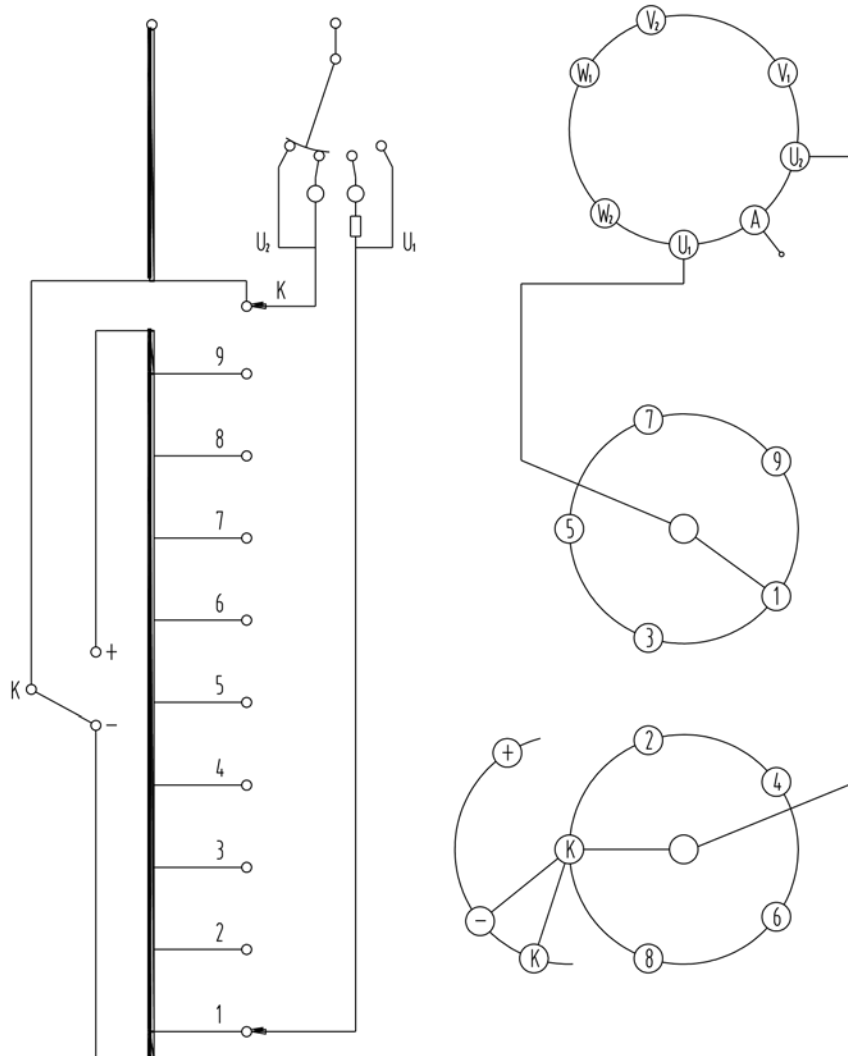


Appendix 9 Dimension of lifting plate for bell type.



Appendix 11 SHGV (10191G) operating position table and connection diagram

diverter switch



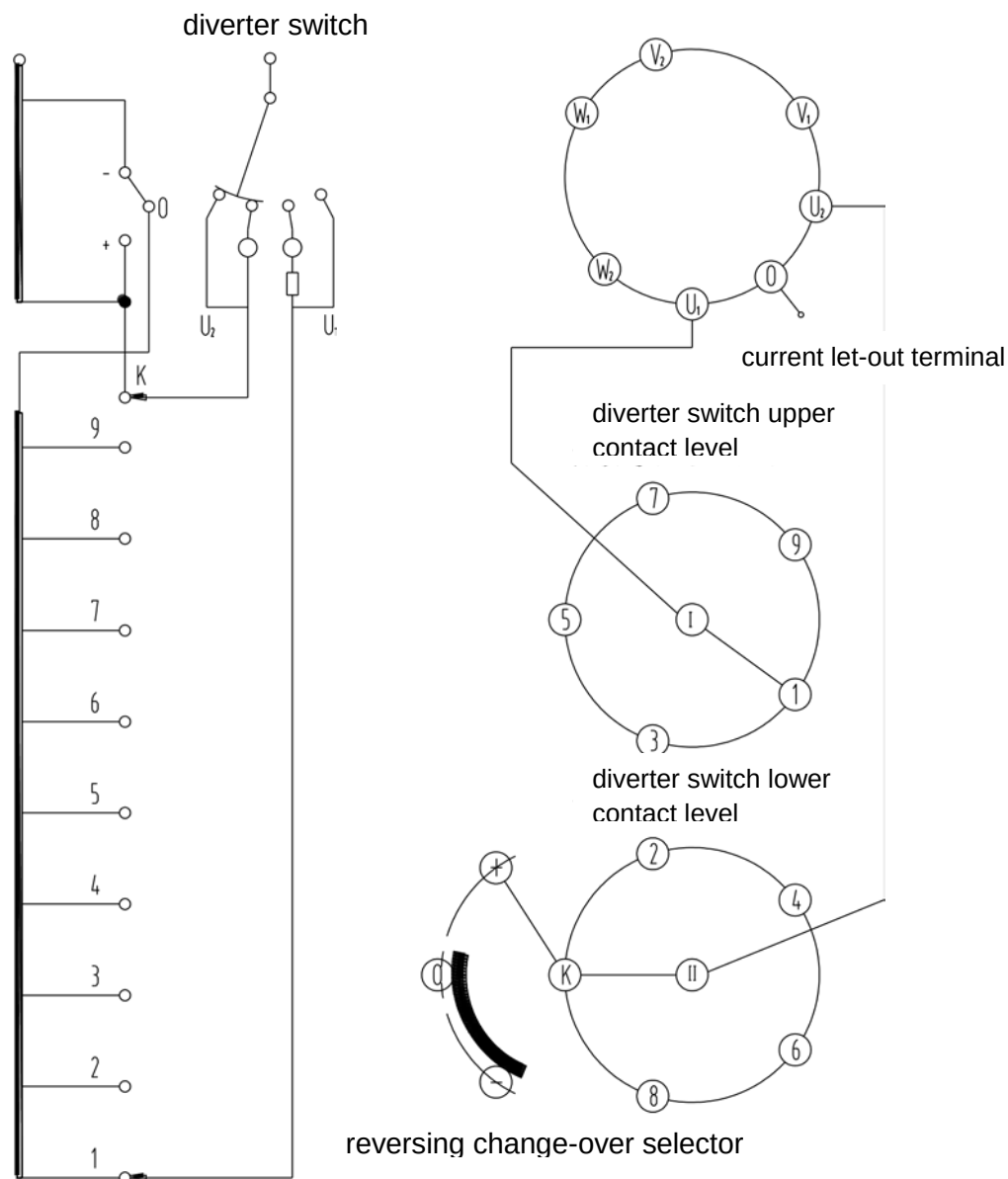
tap changer positions $\blacktriangleleft$	19 $\blacktriangleleft$
different voltages $\blacktriangleleft$	19 $\blacktriangleleft$
setting position $\bullet$ $\blacktriangleleft$	10 $\blacktriangleleft$

change-over position																			
tap selector contact	1	2	3	4	5	6	7	8	9	K	1	2	3	4	5	6	7	8	9
tapping position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

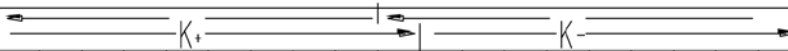


• is setting position

Appendix 12 SHGV (10193W) operating position table and connection diagram



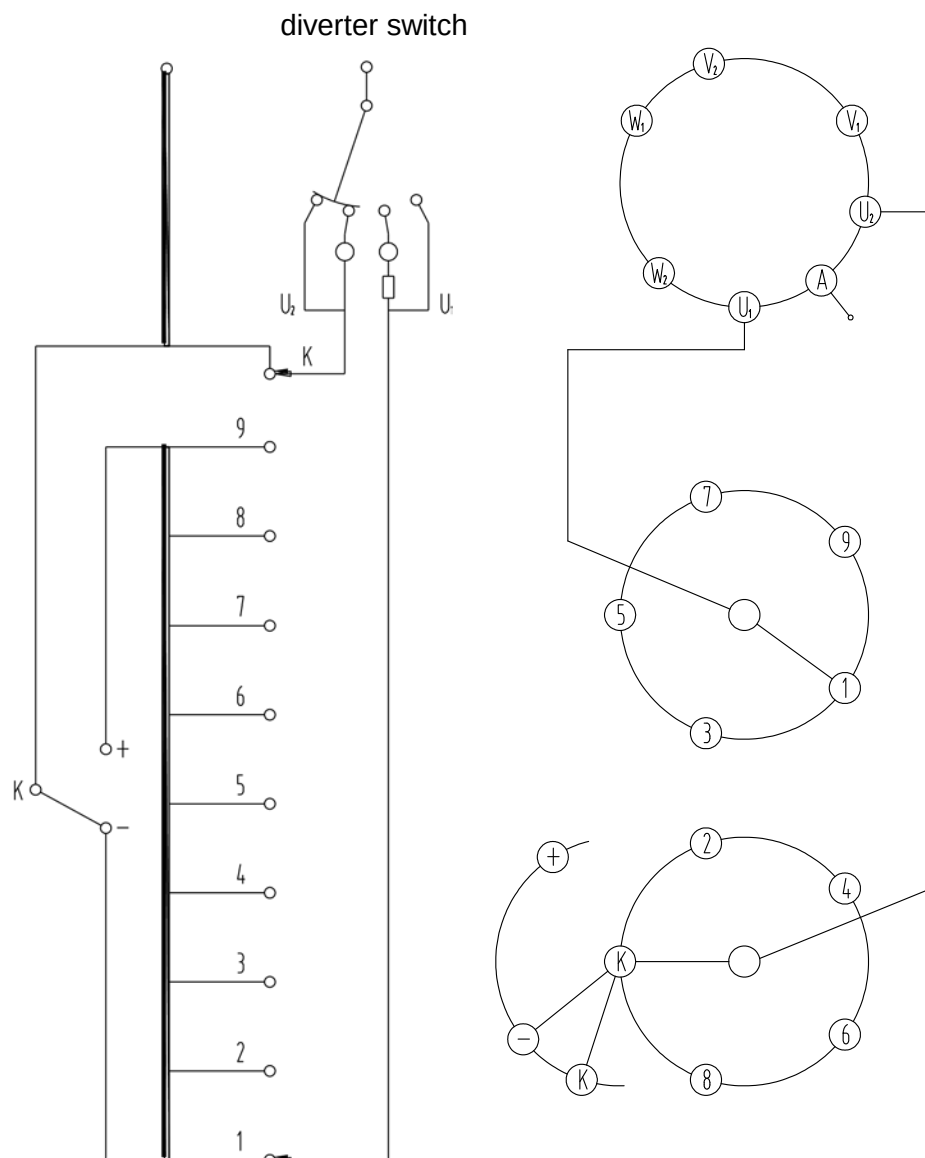
tap changer positions ⁴³	19 ⁴³
different voltages ⁴³	19 ⁴³
setting position ⁴³	10 ⁴³

change-over position																			
tap selector contact	1	2	3	4	5	6	7	8	9	K	1	2	3	4	5	6	7	8	9
tapping position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19



• is setting position

Appendix 13 SHGV (10193G) operating position table and connection diagram

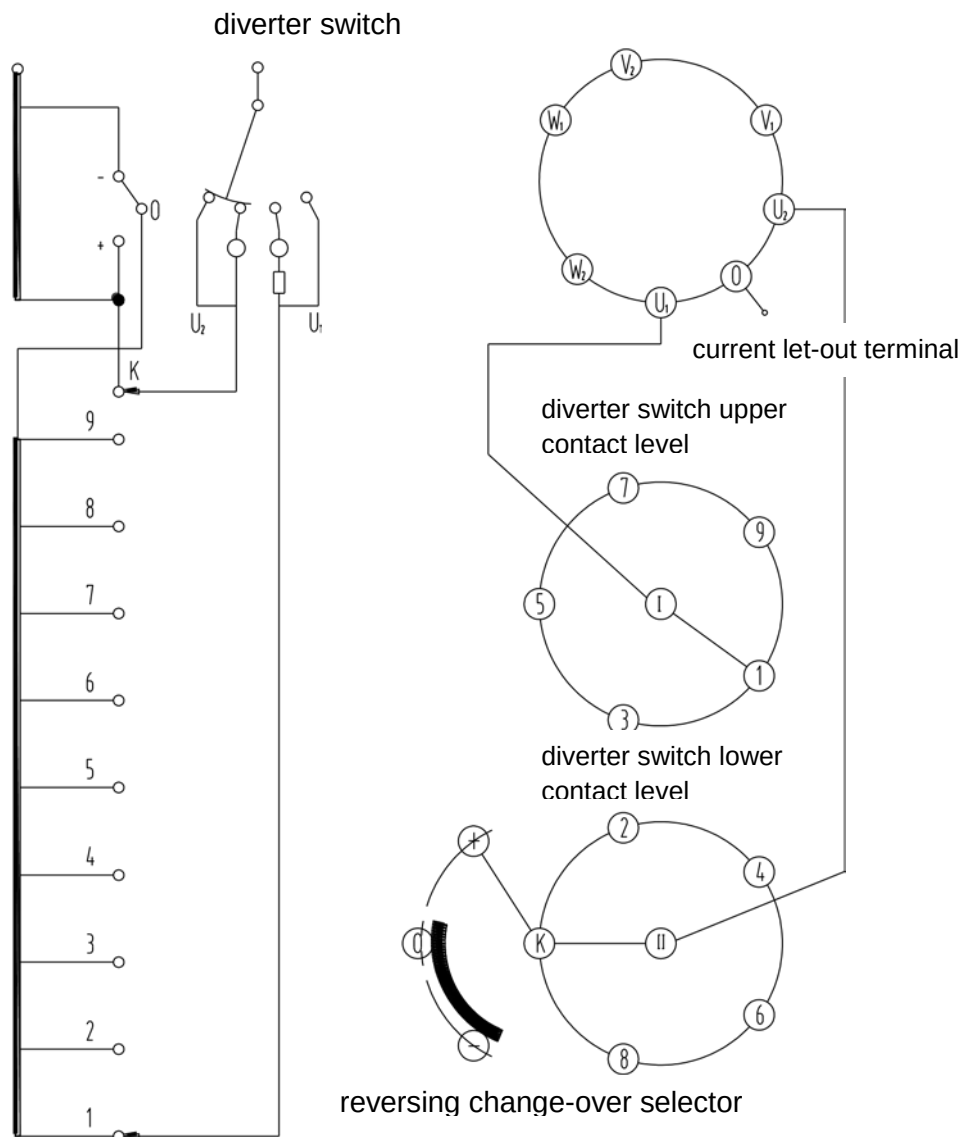


tap changer positions [↕]	19 [↕]
different voltages [↕]	7 [↕]
setting position ● [↕]	9b [↕]

change-over position	<																
----------------------	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

● ← • is setting position

Appendix 14 SHGV (10193G) operating position table and connection diagram



tap changer positions [↕]	19 [↕]
different voltages [↕]	7 [↕]
setting position [•] ↕	9b [↕]

change-over position	</																
----------------------	---	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--



• is setting position

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